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Predicting corporate bankruptcy through a scalable and interpretable knowledge distillation framework

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Abstract

Corporate bankruptcies trigger severe systemic market disruptions. Accurate risk identification remains a critical economic priority. However, existing prediction models struggle with extreme class imbalance. They also face structural noise and high computational costs. This research presents a knowledge distillation framework for bankruptcy prediction. The scope of the present study is strictly limited to structured, tabular financial ratios; integration of unstructured textual data, such as management discussion and analysis and earnings call transcripts, falls outside the purview of this framework and is deferred to future multimodal extensions. The architecture compresses a high-capacity teacher model into a lightweight student network. The methodology utilizes Taiwanese and Polish multivariate datasets. A hybrid mechanism combining LASSO and Mutual Information extracts optimal features. Advanced resampling strategies resolve severe topological class imbalance. These include SMOTE, SMOTEENN, and ADASYN. A blended distillation objective transfers the predictive patterns. This loss formulation mathematically combines temperature-scaled soft probabilities with ground-truth hard labels. The student then employs Histogram Gradient Boosting for rapid inference. Empirical results demonstrate superior predictive fidelity and efficiency against seventeen baselines. On the Taiwanese dataset using SMOTE, the student model achieves 93.77% Accuracy, 69.05% F1, and 93.17% AUC. On the Polish dataset with SMOTE, it delivers 86.79% Accuracy, 63.05% F1, and 88.74% AUC. Across these evaluations, the student model yields merely 0.67% to 1.51% less in F1 score than the teacher model. Crucially, the student model reduces overall computational training time by over 80%. Training consistently completes in under just four seconds. SHAP and LIME algorithms provide essential global and local interpretability. The distilled framework successfully balances predictive accuracy with significantly reduced algorithmic complexity. Ultimately, it offers a scalable, transparent, and highly efficient solution for real-world corporate financial risk assessment.

Keywords Bankruptcy Prediction, Knowledge Distillation, Extreme Class Imbalance, Ensemble Learning, Explainable AI



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1 Introduction

Economic resilience fundamentally relies on financial stability; however, corporate bankruptcies can trigger cascading failures that result in severe systemic market disruptions [1]. One of the main academic priorities is to identify the firms at risk correctly as early as possible [2]. The risk assessment has been led in the past by conventional institutions that are dependent on financial ratios and expert judgment until recently. The exponential growth and increasing complexity of corporate financial data render traditional manual assessment approaches computationally insufficient and practically unsustainable. This restricts the use of traditional heuristic methods and the use of ratios by the analysts. As a result, research focuses on data driven methodologies in order to improve the predictive accuracy and timeliness. These should consider nonstationary, noise and discontinuities in the structure in financial time series. It is therefore imperative to have robust and automatic models of the successful early warning and risk mitigation [3].

Recent innovations in machine learning systems have on the other hand enhanced functionality of predictive modelling in finance field [4, 5]. Accounting data is still valuable in vigorous forecasting attempts with its structured form. It is known that the complementary cues are abundant in unstructured text disclosures [6, 7]. These include management discussion and analysis which articulates futuristic managerial views. These narratives provide insight into what went on with the health of the operations, the risk that lies dormant, and an intention to be strategic beyond the ratios. But even the combination of numeric time series and natural language data introduces unsolved methodological complexity [8, 9]. Heterogeneity of noise, incompatibility across modalities representations and temporal alignment are some of the problems. The consequence of this is that the bankruptcy prediction research has to develop principle and multi-modal fusion techniques. These strategies are to be directed to the scalability, legibility and the ability to stand structural dislocations.

While recent literature demonstrates that complementary predictive cues are abundant in unstructured text disclosures, such as management discussion and analysis (MD&A) [6, 7], it is critical to explicitly note that the scope of the present study is strictly limited to structured, tabular financial ratios. The integration of textual data is beyond the purview of the current framework, allowing us to deeply isolate and resolve the challenges of extreme class imbalance and computational overhead inherent to tabular financial datasets.

The study proposes a novel and superior knowledge distillation model of bankruptcy prediction. An adaptive predictive ensemble teacher model is integrated in the APEX architecture. A low weight DEFT network generates compression of complex patterns into effective frames. Advanced preprocessing methods remove outliers in financial data that are not homogeneous. The ensemble mechanism of feature selection changes either linear or nonlinear predictors of best. Systematic rectification of extreme topological class imbalance Hybrid synthetic sampling strategies. Bayesian meta learner generates good calibrations of target representations. Temperature scaling mechanisms efficiently facilitate the seamless transfer of soft target probabilities from the teacher into the student network. Histogram gradient boosting is useful in accelerating the computational inference and the latency with colossal efficiency. Student model is exceptional fidelity predictive, but with lower complexity. The explainable artificial intelligence tools provide complex global and local interpretability. Experimental findings indicate that

improvements by 17 various baselines are significant. This combined system is effective in solving the major shortcomings of the risk assessment.

It is important to note that while the individual algorithmic components of this framework, such as LASSO, Mutual Information, resampling techniques, and baseline gradient boosting, are established methods, the primary novelty of this research lies in their domain-specific architectural synthesis. Specifically, the APEX-DEFT framework introduces several critical updates tailored for the corporate bankruptcy scenario:

- I. Feature Space Adaptation: The integration of LASSO and Mutual Information within a unified feature selection framework enables the effective reduction of severe multicollinearity and structural noise inherent in high-dimensional financial accounting ratios.
- II. Target Ambiguity Resolution: Knowledge Distillation (KD) is tailored to financial risk prediction through the use of temperature-scaled soft probabilities generated by the teacher model, enabling the transfer of nuanced inter-class relationships within the ambiguous gray zone of corporate distress that are not reflected in conventional hard binary labels.
- III. Latency-Optimized Deployment: Histogram Gradient Boosting Classifier (HGBC) is adopted as the distilled student architecture, leveraging discretized feature binning to mitigate inference latency and computational bottlenecks inherent in real-time institutional credit-scoring environments.

To summarize, the main contributions of this study are as follows:

- I. Systems-Level Integration: APEX-DEFT constitutes a novel end-to-end framework that unifies feature selection, hybrid resampling, and knowledge distillation within a single pipeline designed for tabular financial risk assessment.
- II. Domain-Specific Feature Synthesis: The integration of LASSO and Mutual Information enables the systematic elimination of financial multicollinearity while identifying optimal predictors from high-dimensional tabular data.
- III. Distillation of Financial Risk: Knowledge distillation is adapted to transfer nuanced predictive relationships and soft distress probabilities learned by a stacked generalization teacher into a computationally efficient student model.
- IV. Latency Reduction: The distilled student model achieves over 80% reductions in training and inference time while maintaining near-parity with the predictive fidelity of the teacher model.
- V. Algorithmic Transparency: SHAP and LIME are incorporated into the deployment framework to ensure regulatory-compliant model transparency through complementary local and global interpretability analyses.

The rest of this paper will have the following structure. Section 2 will examine literature of the past and the present approaches to forecasting corporate bankruptcy. Section 3 describes the suggested APEX-DEFT model, which involves pre-processing data, feature selection of the ensemble, removal of class imbalances, and the process of evaluation. Section 4 carries out a comparative study of the experimental results using seventeen different models as the baseline. Section 5 addresses the accountability of the framework in terms of both global and local explainability approaches like SHAP and LIME. Lastly, it presents the Conclusion, which summarizes the empirical findings, highlights

the practical contributions of the study to the assessment of financial risks, and provides the possible prospects of future research.

2 Related work

The prediction of bankruptcy has long been researched on the premise of financial ratio statistical models [1, 2]. Initial models entailed linear mixture of accounting signs in the determination of at risk firms. They gave rules of decisions that would be interpreted yet they frequently could not model intricate patterns. With the progression of time, machine learning methods came into existence to improve precision of prediction [10, 11]. They are such as decision trees, support machine and ensemble. These models are better in dealing with the non-linear and high dimensional input. They are therefore suitable for financial applications in which subtleties of interactions are of importance between variables [12]. As a result, the process of continuing methodological innovation is still imperative for proper forecasting. The main priorities for research should be interpretability, scalability and resilience.

Deep learning approaches were discussed for learning sequential and hierarchical models [13, 14]. Recurrent neural net and convolution models for the analysis of financial time series [15, 16]. They are able to catch dynamic trends which in many cases are not caught by static models. These approaches are better than classical models in terms of short-term forecasting [17]. Fasano et al. [18] proposed a bankruptcy model based on deep recurrent networks. They combined networks with explainable AI; they were able to achieve 0.85 accuracy and 0.927 ROC AUC. They have used 4.1 million records from Orbis from 2012 to 2021. In their model, the researchers processed variable length time series to be better at prediction.

Researchers augment the text information in bankruptcy prediction tasks to improve the accuracy [6, 19]. Unstructured disclosures such as MD&A, earnings calls show strategic outlook and risk [20]. Early work involved a bag of word representations that were semantically poor [21]. The most recent research uses pretrained models for extracting the semantic features in the financial text [7, 22]. Text based models enhance the prediction performance when combined with the sentiment analysis [23]. Lombardo et al. [24] suggested a Transformer system combining accounting data and report text. They used 6190 US Companies spanning from 1999 to 2018 with an accuracy of 87.5%. Their model was based on modeling accounting data with RNNs and summarizing text with an NLP.

Gholampoor et al. [25] used the gradient boosting to forecast bankruptcy on 1,265 firms. They used forty financial ratios and reported the accuracy of classification to be higher than 90%. However, their study is restricted to healthcare and static yearly data of 2022. Consequently, the model has danger of overfitting and has poor generalization over industries and contexts. Taoushianis et al. [26] made use of 1.2 million quarterly records from 1979 to 2023. Their nonlinear logit with fractional polynomials improved Pseudo R² and AUC considerably. It cut Type II errors in half and did not consider unstructured textual disclosures. In contrast our dual encoder is based on feature extraction from structured and unstructured sources.

Mattos et al. [27] examined bankruptcy of 503 distressed private firms in Brazil. Inconsistent accounting made ratios less useful, which increased institutional and legal determinants. Audit status, court specialization, and collateral value proved to be especially strong outcome predictors. Their machine learning models exceeded classic

statistical methods and did not report any accuracy measures. The study did not have architectures for modelling unstructured text, and how to deal with noisy information. Idhmad et al. [28] proposed SVM, Random Forest and Gradient Boosting classifiers. They used 33 financial ratios, governance indicators, and SMOT algorithm in order to solve imbalance. Their best model was 98% accurate but did not include textual data and embeddings.

Despite improvements, few studies in total have deeply examined multimodal learning for the financial risk prediction [8, 9]. Simple fusion schemes such as early or late concatenation are unable to control modality influence [29, 30]. Recent transformer based multimodal frameworks are well connecting tabular, textual and heterogeneous sources. Tabular Text Transformer using cross modal attention by Bonnier et al. [31]. Mathur et al. [32] proposed DocFin, the fusion of tables, reports, and audio of earnings calls. Cao et al. [33] developed RiskLabs, by bringing together market indicators, news sentiment, disclosures. Braithwaite et al. [34] introduced nuFormer with a combination of transaction sequences, textual descriptions for generalization. These transformers enabled multimodal advances are coherent with potential future extensions of our work, although the current APEX-DEFT framework does not implement multimodal fusion and remains focused on tabular financial ratios.

Existing bankruptcy prediction models have large methodological gaps. Traditional statistical methods have not been able to account for complex nonlinear dependencies. Conventional machine learning methods often have problems with extreme imbalance and noise in the structure and different industries generalization. Furthermore, many studies in deep learning have, in addition to the risk of overfitting, an enormous computational overhead, and are missing important algorithmic transparency. In contrast, our proposed APEX-DEFT framework can cross these limitations with an innovative knowledge distillation paradigm. By removing high-compute teacher ensembles into a light weight student network our approach guarantees a very good predictive fidelity FFT and exponentially less computation time. The combination of hybrid resampling approaches ensures that topological skews are robust. Ultimately, having the leverage of SHAP and LIME helps provide full interpretability, making our highly scalable methodology extremely superior in the process of real-world financial risk assessment.

Beyond predictive performance alone, recent advances in financial machine learning have increasingly emphasized the importance of interpretable machine learning. Although complex predictive models often achieve superior accuracy, their opaque decision-making processes introduce substantial regulatory, operational, and governance challenges in high-stakes financial environments. Consequently, contemporary research has focused on integrating advanced predictive architectures with robust interpretability frameworks to generate transparent and actionable insights from financial data. For example, recent work in bank credit customer management [35] combined Extreme Gradient Boosting and Shapley Additive exPlanations with R-learner causal inference to achieve strong predictive performance while identifying the causal influence of key financial behaviors, including transaction frequency and volume fluctuations, on customer attrition. Such studies highlight the growing transition from purely predictive modeling toward explainable and decision-support-oriented financial analytics. Consistent with this methodological direction, the APEX-DEFT framework incorporates both SHAP and LIME within the bankruptcy risk assessment pipeline, enabling

comprehensive global and local interpretability, facilitating transparent model reasoning, supporting regulatory compliance requirements, and enhancing the practical applicability of machine learning-driven bankruptcy prediction in real-world financial decision-making environments.

3 Methodology

This study proposes the highly interpretable knowledge distillation framework for corporate bankruptcy prediction. The model APEX teacher combines well with the lightweight DEFT student network. The range of datasets of Taiwanese's and Polish multivariate bankruptcies are strictly robust scaling and statistical normalized. A novel mechanism that combines LASSO regularization and mutual information for extracting optimal predictive features. Hybrid synthetic sampling strategies systematically coordinate extreme topological skews from extreme class imbalance. Specifically, SMOTE-ENN and ADASYN produce balanced representations in these difficult financial data environments. The APEX teacher has a stacking architecture which uses LightGBM and XGBoost base learners. Thereafter, a Bayesian Ridge classifier meta-learner produces excellent probabilistic representations of target classes. These soft probabilities are passed to the DEFT student through temperature-based distillation loss. The histogram-based DEFT model is sensitive to nuanced topologies and exponentially improves the computational inference latency. To try to make sure that the algorithms are transparent, this framework includes SHAP and LIME to enrich interpretability. Stratified cross-validation is a rigorously-validated predictive efficacy against seventeen machine learning architectures acting as baselines. The entire architectural plan of the proposed APEX-DEFT predictive framework is shown in Fig. 1.

3.1 Dataset description and characteristics

The empirical evaluation of the proposed framework is done on two commonly known and highly imbalanced financial benchmark data sets: the Taiwanese Bankruptcy Prediction dataset and the Polish Companies Bankruptcy data set [36, 37]. Both these datasets are widely used among scholars for research on bankruptcy prediction due to their relevance with practical life, their heterogeneous nature of financial indicators, and their presence of severe class imbalance.

3.1.1 Taiwanese bankruptcy prediction dataset

The database on Taiwanese bankruptcy, which is compiled from Taiwan Economic Journal for 1999–2009, includes the multivariate financial information obtained from firms operating in the regulatory mode of Taiwan Stock Exchange. Under the official business rules of the exchange, a firm is considered bankrupt if it meets the respective bankruptcy criterion, so that the target variable reactivity is a binary variable. The dataset has a total of 6,819 observations and 95 continuous as well as integer-valued financial ratios. It has a drastic class imbalance where there will be around 3.2% of bankrupt cases from the total instances.

3.1.2 Polish companies bankruptcy dataset

The Polish Companies Bankruptcy dataset is a detailed collection of financial and economic indicators for bankruptcy prediction for companies in manufacturing in Poland.

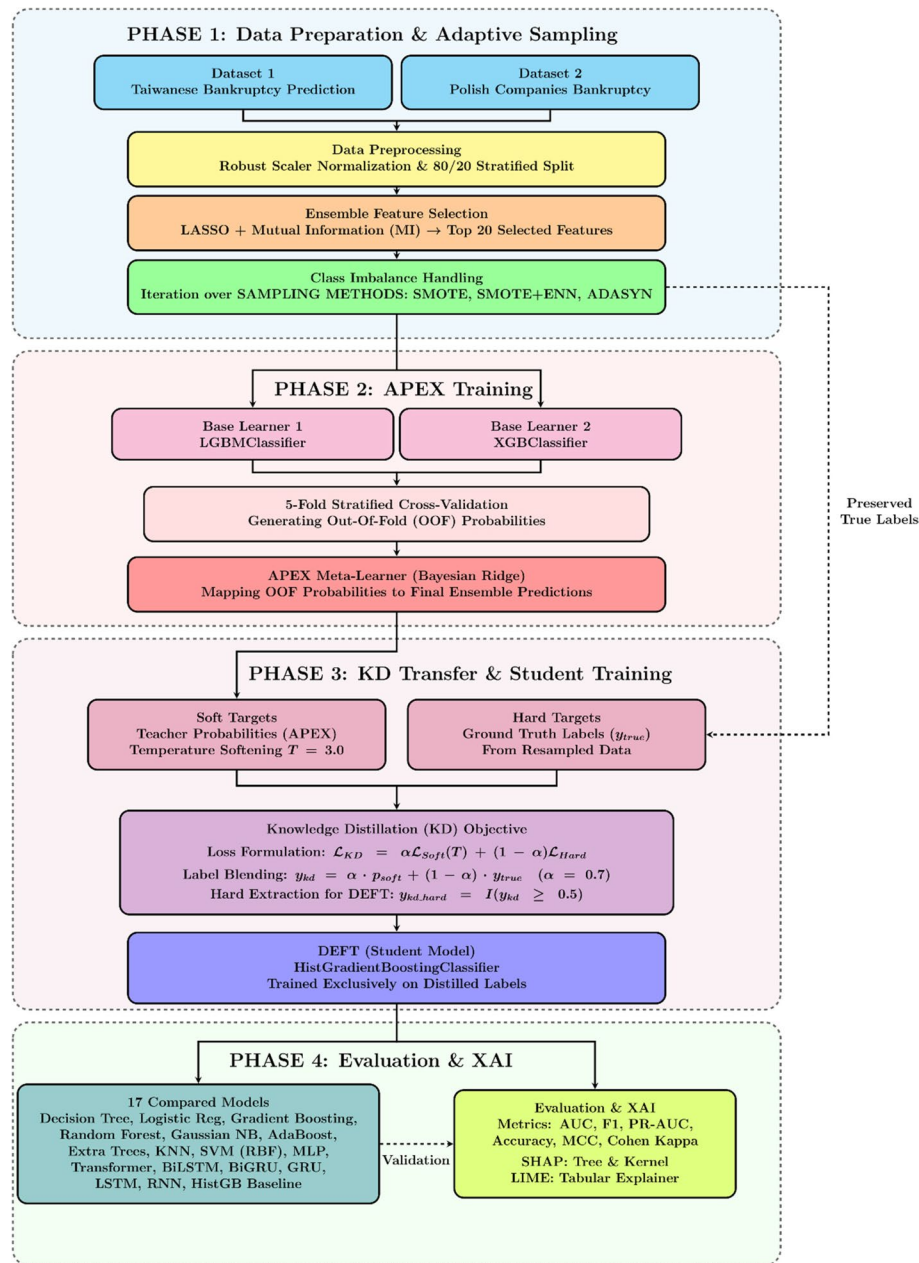


Fig. 1 Architectural Schematic of the Proposed APEX-DEFT Knowledge Distillation and Predictive Framework

The bankrupt companies were studied during the timeframe of 2000–2012 while the non-bankrupt companies that are operating were studied during the period of time of 2007–2013. This data contains 10,503 instances with 65 real-valued financial and economic features. The target variable is a binary variable which distinguishes bankrupt firms from non-bankrupt firms. The dataset is characterized by extreme class imbalance with bankrupt instances representing about 4.8% of the total sample. A topological representation of these extreme class imbalance distributions within both evaluated datasets is visually depicted in Fig. 3.

3.2 Data preprocessing and normalization

These redundant predictors have the potential to adversely affect the further learning algorithms and prediction abilities. Then, feature normalization process is carried out in order to improve the numerical stability for heterogeneous feature distributions. Traditional Z score normalization is very sensitive to extreme data in financial distributions. These extremities are caused by anomalous observations, heavy tailed distributions which are observed. Can thus an effective scaler that takes into consideration median and interquartile range counter actors the effect of outlier effectively. For formal presentation, $X_j \in \mathbb{R}^N$ represents the j -th feature vector. It is the transform of robust scaling and is as follows:

$$X_j^{(scaled)} = \frac{X_j - Q_2(X_j)}{Q_3(X_j) - Q_1(X_j)} \quad (1)$$

where $Q_1(X_j)$, $Q_2(X_j)$, and $Q_3(X_j)$ being the first percentile (25 th percentile), median (50 th percentile), and third percentile (75 th percentile) percentiles of the feature distribution, respectively. This change makes every feature to be centered around its median and scaled by its interquartile dispersion to guarantee toughness to severe financial variability and also to leave the inherent distributional makeup of information.

3.3 Ensemble feature selection: LASSO and mutual information

The level of multicollinearity about huge financial datasets is hindering, as is the degree of feature redundancy, which guarantees that the statistical effectiveness and generalization abilities of predictive models is always diminished. To balance out these weaknesses an ensemble feature selection paradigm is applied to obtain a set of predictors, which are optimally informative and which is represented as S_{opt} . This model is used to combine the linearly regularized data with both the noisy non-linear data theoretic metrics that implant both the dependency of the parameter as well as the non-parameteric dependency in the target variable between the feature space and the target variable.

3.3.1 LASSO regularization (L1 penalty)

The Least Absolute Shrinkage and Selection Operator (LASSO) are one of the coefficient shrinkage and variable selection mechanisms of regularization-based feature selection, which unifies the optimization framework. By introducing a penalty of ℓ_1 -norm on the coefficients of the regression model, LASSO adds a sparsity that actually eliminates irrelevant or less or less important predictors. The optimization objective, usually investigated through cross-validated estimation, is given formally as

$$\min_w \left\{ \frac{1}{2N} \sum_{i=1}^N (y_i - w^\top x_i)^2 + \lambda \|w\|_1 \right\} \quad (2)$$

where $w \in \mathbb{R}^D$ the coefficient vector, x_i the feature vector for the i -th observation, y_i the corresponding target label and λ is the regularization parameter to control the trade-off between the empirical risk minimization and sparsity. Features that are linked to coefficients that shrink to zero are removed in the process and the remaining coefficients represent the linear contribution of the different features. Accordingly, the linear importance score for the j -th feature is set as

$$S_{LASSO}(j) = |w_j| \quad (3)$$

3.3.2 Mutual Information (MI)

LASSO is good at modeling the linear relationships, but it is still restricted in its non-linear relationships. To overcome this limitation, non-parametric measure of dependence - Mutual Information is used. It measures statistical dependence between individual features X_j and the target variable Y . Mutual Information is a measure of reduction in uncertainty of one variable by knowing about the other. This property makes them especially suitable for the identification of nonlinear and non-monotonic relations. Formally, mutual information is the Kullback-Leibler divergence between joint probability distribution and marginal ones.

$$I(X_j; Y) = \int_Y \int_{X_j} p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right) dx dy \quad (4)$$

where $p(x, y)$ denotes the joint probability density function, and $p(x)$ and $p(y)$ represent the marginal densities. In practice, this quantity is estimated using k-nearest neighbor-based entropy estimation techniques to ensure robustness in high-dimensional and non-Gaussian settings. The resulting value serves as the nonlinear importance score, denoted as $S_{MI(j)}$.

3.3.3 Information fusion

In order to build a coherent importance measure which captures the linear sparsity-driven relevance, simultaneously with nonlinear dependency structure, the two score vectors are normalized on a common scale. Specifically, both S_{LASSO} and S_{MI} undergo the min-max normalisation, and the normalised vectors $\hat{S}_{LASSO}$ and $\hat{S}_{MI}$ can be obtained, lying respectively in the interval $[0, 1]$. The importance score of the ensemble for the j -th feature is then defined as the arithmetic mean of the normalized scores:

$$S_{Ensemble}(j) = \frac{\hat{S}_{LASSO}(j) + \hat{S}_{MI}(j)}{2} \quad (5)$$

This combination ensures the matching of the opposite statistical views that consequently enhance the strength in the estimation of the feature relevance. The final feature subset is selected by selecting the optimal K features that are computed based on $S_{Ensemble}(j)$, where $K=20$. This dimensionality reduction algorithm considerably relieves redundancy and noise and reveals the most salient predictive signals, which eventually enable better model interpretability, as well as the generalization model ability. Importance using these hybrid mechanisms is ranked in Fig. 2.

The specific threshold of $K = 20$ was deliberately established based on preliminary empirical cumulative importance analyses. In highly dimensional financial datasets, accounting ratios frequently exhibit severe multicollinearity, meaning that a vast majority of features provide highly redundant economic signals. Our preliminary evaluations revealed that the top 20 features capture the overwhelming majority of the total predictive variance. Extending the feature subset beyond this threshold introduced strict diminishing returns, exponentially increasing computational overhead and structural noise without yielding statistically significant improvements in out-of-fold F1 or AUC

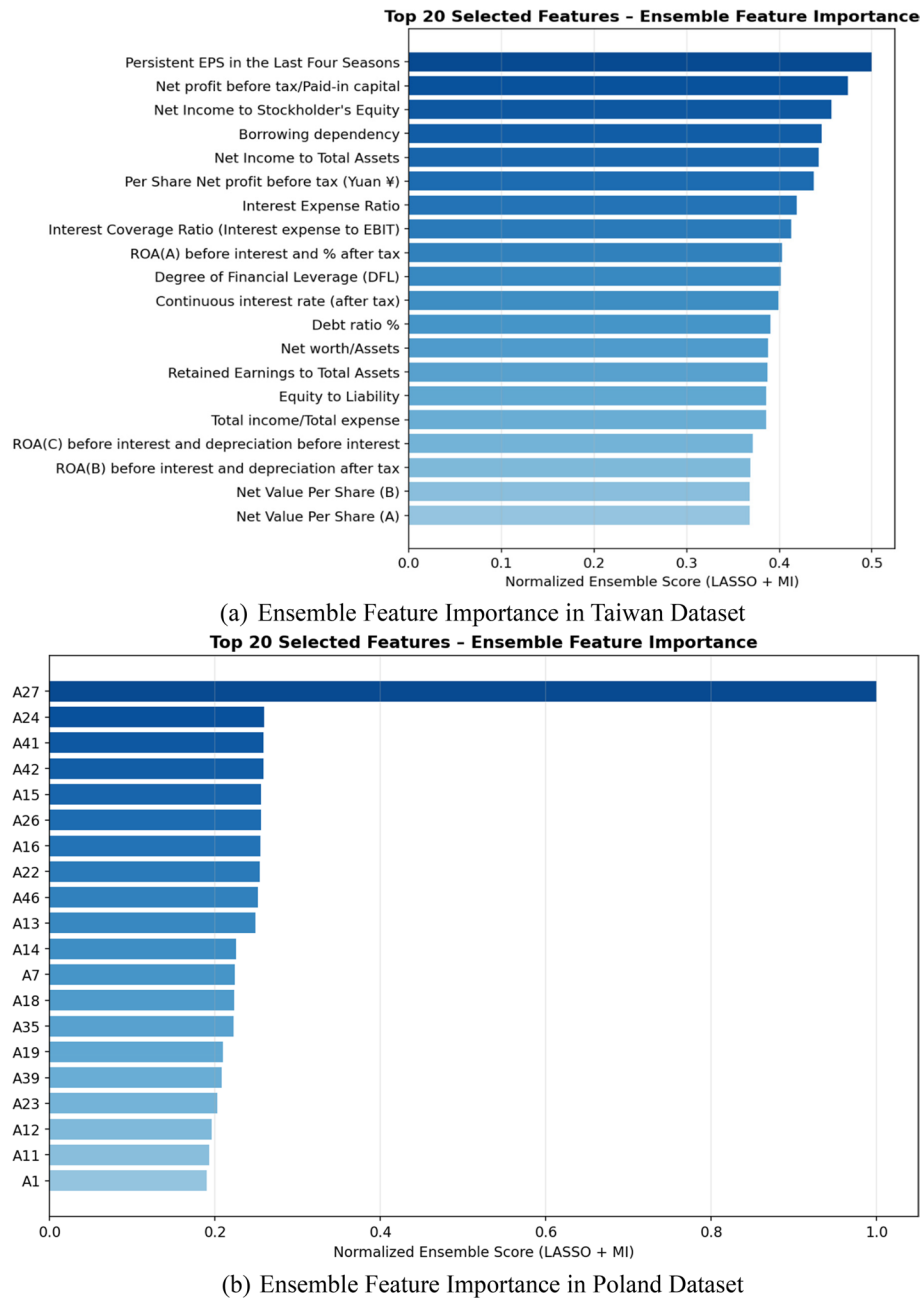


Fig. 2 Ranked Ensemble Feature Importance Derived from Hybrid LASSO and Mutual Information Mechanisms

scores. Furthermore, restricting the optimal subset to exactly 20 predictors ensures that the subsequent SHAP and LIME global interpretability diagnostics remain cognitively tractable and practically actionable for human risk analysts. This dimensionality reduction algorithm considerably relieves redundancy and noise and reveals the most salient predictive signals, which eventually enable better model interpretability, as well as the generalization model ability.

3.4 Class imbalance mitigation strategies

Extreme class imbalance significantly skews supervised learning algorithms, disproportionately biasing decision boundaries toward the majority class and hindering the

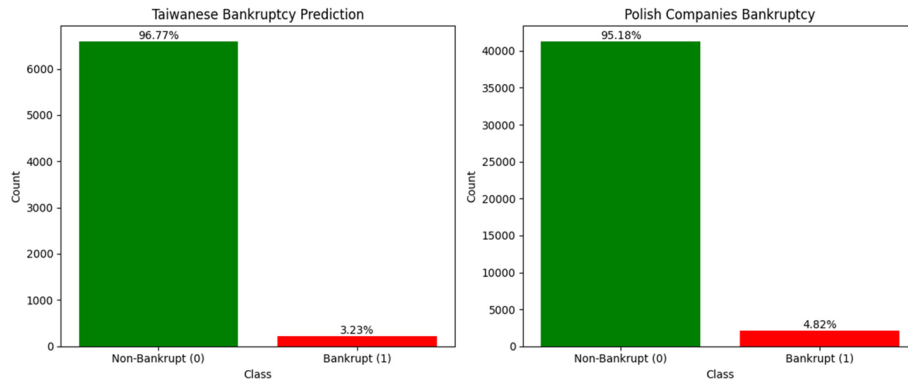


Fig. 3 Topological Representation of Extreme Class Imbalance Distributions within the Evaluated Financial Datasets

detection of minority instances. This bias affects actually in favour of the non-bankrupt companies and lessens the detection of the minor bankrupt cases. Consequently, there is a reduction in model sensitivity and systematically poor recall for financially distressed observations. To overcome this drawback there are sophisticated resampling techniques to reconstruct a better-balanced training distribution. These strategies increase the classifier ability on detecting rare but economically important bankruptcy occurrences. First, SMOTE comes up with the idea to introduce synthetic oversampling, where interpolation of minority samples with nearest neighbors is performed. This process creates new instances of minorities making representation in feature space better.

$$x_{new} = x_i + \delta \cdot (x_{zi} - x_i), \delta \sim Uniform(0, 1) \quad (6)$$

Here, δ represents a random scalar uniform variable with range 0–1. This interpolation fills up feature space between line segments between minority samples effectively. It increases the minority class support but decreases the sparsity and does not duplicate existing observations excessively.

ADASYN extends the oversampling technique by giving the synthetic samples more focus toward the more difficult minority instances. These instances lie in regions that are populated with majority class neighbors, making it more difficult to classify these instances. A normalized difficulty distribution is a measure of relative complexity that influences the generation of the synthetic samples in proportion. In addition, SMOTE ENN also combines oversampling with neighborhood-based cleaning to eliminate misclassified samples.

$$y_i \neq mode(\{y_j : x_j \in N_k(x_i)\}) \quad (7)$$

In this case, $N_k(x_i)$ would be nearest neighbors, and mode would be the most popular class label. This facilitates the elimination of noise of borderline samples, assists to smooth the boundaries of classes in order to enhance the integrity of data. All these measures intend to enhance representation of minorities and prevent spurious observations.

3.5 Proposed architecture: the APEX teacher framework

In the suggested methodological framework, it is possible to consider that the extraction and combinations of high-dimensional predictive representations are in fact around the hierarchical ensemble paradigm. Adaptive Predictive Ensemble eXpert (APEX)

manifests itself as a stacked generalization architecture; in this framework a stack of heterogeneous base learners is ensembled by an ensemble between them; that is, a meta-learning mechanism thereof. This framework exploits the cross-validation predictions based on the Out-of-Fold (OOF) predictions to create an information-rich and unbiased meta-feature space and hence overcome the issue of overfitting without the sacrifice of the generalization ability.

3.5.1 Base learners: *lightGBM and XGBoost*

At the first stacking level, two advanced models of the Gradient Boosting Decision Trees are used. These are LightGBM and XGBoost that are selected because of good nonlinear modelling and robust. The two algorithms have good empirical results in structural heterogeneous tabular financial data. LightGBM is a leaf wise growing algorithm and expands leaves that have maximum reduction in the objective function. This enables more depth of trees that provide more representational ability of sophisticated interactions of features. It also takes into consideration Gradient based One Side Sampling and Exclusive Feature Bundling optimization. Functional approximation on the objective in iteration t is calculated by implementing second order Taylor expansion.

$$Obj(t) \approx \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \quad (8)$$

In this case, g_i and h_i represent first and second order gradients of loss function. Model complexity is regulated by the term $\Omega(f_t)$: regularization is obtained through the explicit regularization of the model when optimizing it. On the other hand, XGBoost takes the level wise growth, which guarantees the balancing of trees and better computational power.

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (9)$$

Here, T represents number of leaves, w is leaf weights and γ and λ are regularization of complexity. This formulation addresses the risks of overfits within out of fold trainings effectively. Both learners use the class sensitive weighting that puts more emphasis on the minority misclassification penalty and enhances boundary discrimination.

3.5.2 Meta-learner: *bayesian ridge classification formulation*

The second stacking level is based on a meta feature space based on the OOF predictions. Outputs of LightGBM and XGBoost in P_{LGBM} and P_{XGB} are concatenated respectively. This creates a matrix Z , which is fed into the meta learner for the purpose of classifying it. In order to get robust mapping, Bayesian Ridge Regression in a probabilistic classification application is adopted. This uses Gaussian priors of regression coefficients leading to principled regularization and stability. Likelihood of measurements vs. Reliability in Gaussian Formulation This prediction formulation offers further support to uncertainty quantification for predictive modeling tasks. Consequently, posterior weight distributions are calculated by Bayesian theorem for probabilistic inference.

$$p(w \mid Z, y, \alpha, \lambda) \propto p(y \mid Z, w, \lambda) p(w \mid \alpha) \quad (10)$$

Here, α and λ represent precision parameters of prior and likelihood distributions. The model makes continuous predictions not enclosed in the bounds of probability. Inverse logit transform an output to get calendar posterior quickness

$$P(y = 1 \mid z) = \sigma(\hat{y}_{raw}) = \frac{1}{1 + \exp(-\text{clip}(\hat{y}_{raw}, -500, 500))} \quad (11)$$

Clipping helps in the numerical stability by capping the numerical values of extreme values and function σ stands for sigmoid logistic function. This mapping is what makes the application of a probabilistic interpretation possible. The calibrated Bayesian meta learner will give final predictive signal, combining the boosting and the probabilistic modeling strengths.

3.6 Knowledge Distillation (KD) mechanism

Knowledge Distillation is a principled model compression paradigm, which is expected to transfer the predictive competence of a high-capacity, computationally-intensive model into a considerably more efficient surrogate model. And one of the roles of the APEX Teacher is to be an enriched source of supervisory signals, allowing the Student model not only to learn explicit decision boundaries but also to lean on the latent structure information packed in the probabilistic outputs of the teacher. This approach is much more than supervised learning, since it involves recommending soft target distributions that reflect subtle inter-class relationships.

3.6.1 Temperature scaling and soft target generation

Let z_T denote the logit-level outputs of the teacher model prior to the application of the final sigmoid transformation. To recover informative soft targets a temperature scaling mechanism is added, which is determined by a hyperparameter $T > 0$. This parameter modifies how smooth the resulting probability distribution will be, and higher values will lead to the resulting probability distribution being softer and/or more informative than distributions that are obtained by using only the individual class probabilities. The softened probability q_i for class i is calculated by applying a temperature controlled Soft-max function:

$$q_i = \frac{\exp(z_{T,i}/T)}{\sum_{k \in \{0,1\}} \exp(z_{T,k}/T)} \quad (12)$$

For elevated temperature values, such as $T = 3.0$, the output distribution spread less, therefore, exposing the relative confidence of the teacher across classes. This softened distribution packages what can be called as the “dark knowledge” in terms of subtle inter-class similarities, the decision uncertainties which cannot be transduced by some hard binary labels alone. Such a fact is especially useful in unbalanced classification problems, where minority class distinctions may be inherently ambiguous.

3.6.2 Distillation loss function

The Student model is trained using the supervisory signal which is a convex combination of the teacher softened probabilities and the ground truth binary labels. It is a

mixed target which simplifies the transfer of the global class arrangement and the predictive patterns in the fine grain. The formal target of distillation, y_{KD} is defined as

$$y_{KD} = \alpha q + (1 - \alpha) y_{true} \quad (13)$$

where q denotes the vector of softened class probabilities produced by the teacher, $y_{true} \in \{0, 1\}$ represents the ground-truth label, and $\alpha \in [0, 1]$ is an interpolation coefficient controlling the relative contribution of the soft and hard targets. A value of $\alpha = 0.7$ assigns greater emphasis to the teacher-derived knowledge while retaining alignment with the true labels.

The Student model is then trained by rewriting the student model to minimize the cross-entropy divergence between the student predicted probabilities and K stable target y_{KD} . This objective function guarantees that the Student learns both the macroscopic class discrimination contained in the ground-truth labels and the microscopic relational structure which is present in the teacher's probabilistic outputs. As a consequence, the distilled model attains a favorable balance between computational efficiency as well as predictive fidelity and approximates the performance characteristics of the original APEX Teacher under greatly diminished complexity.

3.7 The DEFT student architecture

The Distilled Efficient Financial Threat detector (DEFT) is initialized using the Histogram-based Gradient Boosting Classifier (HGBC), which is a computationally optimized version of the gradient boosting for large-scale tabular data. In contrast to more standard forms of decision tree learners that use input data directly on continuous feature spaces, HGBC provides a discretization mechanism where continuous features are subdivided into non-overlapping integer value bins that are usually in the order of 256 different bins each. This revolution is a radical re-arrangement of the computational organization of the learning process and does tremendous efficiency acquisitions without significantly impairing the fidelity of the predictions under consideration.

Let $x_j \in \mathbb{R}$ symbolically represent a continuous feature. The binning is used in which the x_j goes to a discrete representation of $b_j \in \{1, 2, \dots, B\}$, where $B \ll N$ is the size of the training instances. By doing this discretization the complexity of split finding is reduced when constructing a tree. In contrast to the traditional algorithms, in which each readability or sorting operation is limited to computational effort of $O(N \log_N)$, under the precondition of histogram input technique, its computation effort could be under $O(B)$. This decrease is particularly significant in the high dimensional and large sample regimes which is the only one that abates in terms of computational scalability. The HGBC underlying optimization problem is similar to gradient boosting in which a sequence of additive tree models are induced by minimizing a regularized loss function. At iteration t , the objective can be written as

$$Obj^{(t)} = \sum_{i=1}^N L(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \lambda \|f_t\|_2 \quad (14)$$

where $L(\cdot)$ is a loss function, f_t is the tree inserted at iteration t , and λ is the regularization parameter which controls the complexity of the model. When using the DEFT

architecture, a ℓ_2 -regularization penalty is applied to restrict the scale of the model parameters, which reduces the chances of overfitting to the distilled supervisory signals.

Another such salient strength of HGBC is that the missing value processing is part of it. Missing entries are equivalent to separate bins or optimal routing while split evaluation is going on. This eliminates the need to explicitly specify the imputation requirements and at the same time ensures that the data remains intact when being trained and inferred. The representation of histogram also enables effective inference using the few integer comparisons, which significantly minimizes the time taken to compute. To make a better generalization, it is the internal validation subset that is used to implement the early stopping. The training halts when the validation performance ceases to increase to prevent overfitting to soft targets. All these mechanisms enable the DEFT Student architecture to be highly efficient and scalable, showing strong potential for real-time applications, though further evaluation of high-concurrency throughput and incremental updating is needed.

3.8 Theoretical formulation of the 17 baseline models

In order to demonstrate the usefulness of the APEX-DEFT knowledge distillation framework in concrete way, a wide-scale comparative analysis is established with a variety of seventeen baseline models of various branches the classical statistical learners, ensemble learning models and advanced neural networks. The general character of this benchmarking plan ensures that the suggested framework will be considered in the broad spectrum of modeling paradigms of alternative inductive biases, representational possessions and optimization processes.

3.8.1 Classical and ensemble machine learning baselines

Logistic Regression is a good example of a base linear classifier, in which a posteriori probability of the binary target variable is modeled by a logistic transformation of a linear predictor. The likelihood of a bankruptcy is given by

$$P(Y = 1 \mid X) = \frac{1}{1 + \exp(-(\beta_0 + \beta^\top X))} \quad (15)$$

where β denotes the coefficient vector and β_0 is the intercept term. The number parameters of the model are estimated using maximum likelihood by minimizing the cross-entropy loss. Decision Trees are a non-parametric model, in which, the feature space is recursively divided into increasingly homogeneous regions. At each split, some impurity measure such as the Gini index,

$$Gini = \sum_{i=1}^C p_i(1 - p_i) \quad (16)$$

Information gain calculated from reduction of entropy is optimized to find the best partitioning criteria. Random Forest is an extension of the decision tree concept where bootstrap data aggregation is used to build decision tree ensembles with data subsets from resampling. Final predictions are got through averaging which boosts stability and reduces variance.

$$\hat{f}_{RF}(x) = \frac{1}{M} \sum_{m=1}^M \hat{f}_m(x) \quad (17)$$

This process has the advantage of reducing the variance and increasing generalization under a variety of unseen data distributions. Gradient Boosting works as forward stage wise additive model of empirical risk minimization. In successive iterations of weak learners, fit negative gradients, correcting residual errors in the previous iterations. Hist-GradientBoosting adds this extra efficiency by breaking continuous features into small histogram bins. This allows for approximate split finding with a high level of preservation of predictive performance using different datasets. However, this baseline does not include technique for knowledge distillation although there is structural similarity to DEFT architecture. Extremely Randomized Trees use random rather than deterministic choices of split thresholds. AdaBoost: A sample re-weighting is iteratively employed to give a focus to the mis-classified instances using adaptive weight update mechanism.

$$w_i^{(m+1)} = w_i^{(m)} \exp(\alpha_m I(y_i \neq G_m(x_i))) \quad (18)$$

Here, α_m denotes learner weight, while I represent the indicator function. Gaussian Naive Bayes is more a conditional independence while modeling features using a Gaussian distribution. K Nearest Neighbors diagnosis by adventure majority voting among closest observations. Using radial basis function of kernels to create maximum margin hyperplanes.

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (19)$$

where γ controls the kernel width. Optimization is performed in the dual space, commonly via Sequential Minimal Optimization.

3.8.2 Deep learning and recurrent baselines

The inclusion of several neural architectures is undertaken to give context of the performance in representation learning. These models use adapted gradient optimization or dropout regularization to reduce the effects of overfitting. Multi-Layered Perceptron Multi-Layered Perceptron is a DL with non-linear activation functions. Rectified Linear Units introduce Nonlinearity allowing to learn complex feature representations well. The output layer is using the sigmoid or softmax activation function for binary classification problems. Recurrent Neural Networks represent sequential dependencies in the form of changing representations of hidden states over time

$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b_h) \quad (20)$$

where h_t is the hidden state at time t . Long Short-Term Memory networks eliminate the vanishing gradient problem by incorporating gated mechanisms and controlling the information flow. The dynamics of LSTM cell are dictated by.

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) \\ \tilde{c}_t &= \tanh(W_c[h_{t-1}, x_t] + b_c) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \\ h_t &= o_t \odot \tanh(c_t) \end{aligned} \quad (21)$$

where f_t , i_t , and o_t denote the forget, input, and output gates, respectively. Gated Recurrent Units are a computationally efficient alternative, which combines gating mechanisms. The equations or updates are given by

$$z_t = \sigma(W_z[h_{t-1}, x_t]) \quad (22)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot h_t^?$$

Here, z_t represents the update gate on the flow of information in recurrent units. Bi directional LSTM and GRU process the sequences in both ways i.e. forward and backward, while helping us to understand more context dependency. Transformer models abandon recurrence in favor of self attention and, therefore, are able to directly model global interactions of features.

$$Attention(Q, K, V) = softmax\left(\frac{QK^\top}{d_k}\right)V \quad (23)$$

where Q, K, and V denote the query, key, and value matrices, respectively, and d_k is the dimensionality of the key vectors. This scaling factor stabilizes propagation of the gradient and preserves numerical consistency in optimizing. Together with these baseline models, a rough and theoretically valid standard is provided on which a rigorous assessment of the suggested framework of APEX-DEFT regarding various paradigms in the field of methodology could be conducted so far.

3.9 Evaluation protocols and performance metrics

In the case of extremely skewed classification, traditional evaluation metrics, such as overall accuracy, are notoriously misleading because they overwhelmingly account for the majority class and hide the model's ability to identify minority instances of critical importance. This phenomenon is known as the accuracy paradox and requires more discriminative and imbalance-sensitive criteria for evaluation. Accordingly, the predictive performance of all the baseline models along with the architectures APEX Teacher and DEFT Student is evaluated under a Stratified 5-fold Cross Validation protocol, both of which ensure that the class distribution is maintained across all folds while maintaining evaluation statistical consistency. The scope of the evaluation system focuses both on class-specific precision-recall summary and threshold-independent ranking evaluation.

To strictly prevent data leakage and ensure the integrity of the empirical evaluation, all class imbalance mitigation techniques, including SMOTE, SMOTE-ENN, and ADASYN, were applied exclusively to the training subsets within each cross-validation iteration. The validation folds were preserved in their original, highly imbalanced state without any modification. This isolation protocol ensures that the student model is assessed on authentic out-of-sample distributions, thereby providing an unbiased and realistic estimate of its generalization capability in real-world financial environments.

$$Precision = \frac{TP}{TP + FP} \quad (24)$$

where TP and FP denote true positives and false positives, respectively. Complementarily, recall measures the completeness of the classifier in capturing all relevant positive instances and is expressed as

$$Recall = \frac{TP}{TP + FN} \quad (25)$$

where FN represents false negatives. To balance these two competing objectives, the F1-score is employed as their harmonic mean:

$$F_1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (26)$$

All the metrics use macro averaging to ensure the same weighting to each class. This method helps to overcome the bias caused by severe class imbalance when conducting evaluation. ROC AUC measures discrimination ability, for all the possible decision thresholds, effectively. It is the integral of the ROC curve of the true and false positive rates. However, ROC AUC may overestimate performance in cases where data is imbalanced. Therefore, PR AUC takes into account to better represent minority class predictive performance. Additionally, the strength of Matthews Correlation Coefficient is comprehensive evaluation using the confusion matrix.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (27)$$

Here, TN is true negatives to guarantee that it is evaluated comprehensively under extreme class imbalance. This measure coincides with Pearson correlation which ensures good assessment reliability of binary classification. Cohen Kappa is another measure of predicted and true agreement in excess of chance. In addition, computational efficiency is also measured using training and inference time analysis.

4 Comparative performance analysis

4.1 Framework overview and baseline comparison scope

The proposed knowledge distillation framework establishes a comparative benchmark with seventeen different predictive models. This is an integrated assessment of the classical ensemble; as well as concepts of modern deep learning models. The main aim is to give a quantitative account of the mechanics of the trade-off between fidelity of predictions made in terms of the power to predict new observations and practicality in terms of the time of interaction. APEX is the high-power generator of complicated probability distributions of the teacher. DEFT is the student who is the compressor of light weight of this distilled knowledge. Comparative analysis is conducted across both datasets of Taiwanese and Polish bankruptcy under consistent experimental protocols. Stratified 5-fold cross validation is employed to guarantee the statistical robustness of all reported metrics. Performance evaluation emphasizes minority class recall and threshold independent measurements due to innate data imbalance.

4.2 Sampling strategy impact on teacher-student performance

Sampling techniques have a huge effect on classification results in the case of severe class imbalance conditions. Table 1 outlines the performance of APEX and DEFT for

Table 1 Performance Comparison of APEX Teacher and DEFT Student Models on Taiwan Dataset Using Different Sampling Techniques

Sampling	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	AUC (%)	PR AUC (%)	MCC (%)	Cohen Kappa (%)	Train Time (sec)
SMOTE	APEX (Teacher)	94.40 ± 0.48	65.74 ± 1.39	81.73 ± 1.33	70.56 ± 1.58	90.49 ± 0.89	52.88 ± 3.30	44.68 ± 2.88	41.51 ± 3.08	10.27 ± 0.24
	DEFT (Student)	93.77 ± 0.15	64.30 ± 0.39	81.40 ± 0.81	69.05 ± 0.50	93.17 ± 0.36	50.75 ± 2.61	42.38 ± 1.05	38.62 ± 0.99	1.55 ± 0.05
SMOTEENN	APEX (Teacher)	91.32 ± 0.20	61.67 ± 0.33	85.19 ± 0.68	66.12 ± 0.47	90.68 ± 0.81	39.54 ± 3.27	40.52 ± 0.94	33.61 ± 0.89	9.42 ± 0.08
	DEFT (Student)	90.51 ± 0.40	61.03 ± 0.39	85.87 ± 0.59	65.20 ± 0.58	93.18 ± 0.41	42.49 ± 2.82	39.78 ± 0.88	32.09 ± 1.02	1.40 ± 0.12
ADASYN	APEX (Teacher)	94.11 ± 0.45	64.91 ± 1.13	80.92 ± 0.79	69.60 ± 1.25	90.16 ± 1.19	51.41 ± 3.20	42.93 ± 2.12	39.64 ± 2.40	10.30 ± 0.12
	DEFT (Student)	93.34 ± 0.33	63.55 ± 0.72	81.40 ± 1.02	68.22 ± 0.89	92.97 ± 0.31	51.11 ± 1.18	41.26 ± 1.67	37.09 ± 1.72	1.58 ± 0.04

three resampling strategies for the Taiwan dataset. SMOTE achieves better precision by the teacher is 65.74% and good recall is 81.73%. DEFT has a comparable performance of getting 64.30% of precision and getting 81.40% of recall. SMOTE-ENN increases teacher recall to 85.19% and decreases precision to 61.67%. ADASYN delivers a balance between teacher metrics with precision 64.91% and recall 80.92%. Training time reduction by teacher to student total > 80% used in all sampling strategies. Furthermore, the comparative confusion matrix diagnostics illustrating the predictive efficacy between the APEX Teacher and DEFT Student architectures are detailed in Fig. 4.

4.3 Cross-validation stability across sampling strategies

Table 2 shows similar comparisons between performances on the Poland data set showing that consistent patterns emerge. SMOTE-ENN gives the best result in teacher recall (79.21%) with a precision level of 58.45%. DEFT demonstrates tremendous fidelity that keeps recall and precision at 79.19% and 58.52% respectively. ADASYN comes up with precision of 60.15% by teachers and recall of 77.60%. Student models suffer negligible degradation in performance in all aspects. Training time compression still remains large with student models completing time in 4 s. The Poland dataset has lower general metrics indicating the higher complexity.

4.4 Fold-level performance consistency for Taiwan dataset

It's granular results from 5-fold cross validation for Taiwan dataset is given in Table 3. APEX shows stable teacher performances in all five folds. SMOTE achieves teacher F1 slide from 68.96% to 72.74%. DEFT gets a similar F1 score of 68.47% – 69.61%. To improve the teacher recall, SMOTE-ENN reaches 85.76% in the third fold. Student recall of the course is still high at 86.52% in the fifth fold. ADASYN is showing teacher F1 scores ranging from 68.16% to 71.53%. Training time for DEFT is always less than 1.7 s/fold. This stability proves the strength of knowledge distillation on the sampling strategies.

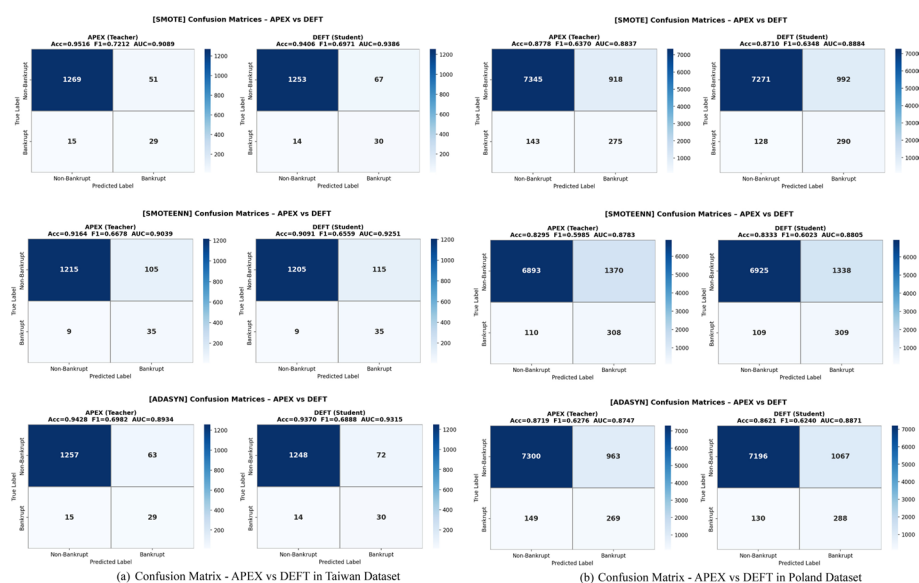


Fig. 4 Comparative Confusion Matrix Diagnostics Illustrating Predictive Efficacy Between APEX Teacher and DEFT Student Architectures

Table 2 Performance Comparison of APEX Teacher and DEFT Student Models on Poland Dataset Using Different Sampling Techniques

Sampling	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	AUC (%)	PR AUC (%)	MCC (%)	Cohen Kappa (%)	Train Time (sec)
SMOTE	APEX (Teacher)	87.70 ± 0.05	60.59 ± 0.17	77.62 ± 0.59	63.72 ± 0.24	88.15 ± 0.17	42.81 ± 0.59	34.20 ± 0.64	29.16 ± 0.50	28.15 ± 0.11
	DEFT (Student)	86.79 ± 0.21	60.15 ± 0.15	78.28 ± 0.55	63.05 ± 0.22	88.74 ± 0.16	41.17 ± 0.72	33.88 ± 0.51	28.19 ± 0.41	3.89 ± 0.05
SMOTEENN	APEX (Teacher)	82.70 ± 0.20	58.45 ± 0.13	79.21 ± 0.34	59.84 ± 0.22	87.89 ± 0.10	34.00 ± 0.53	31.42 ± 0.41	23.64 ± 0.36	25.59 ± 0.13
	DEFT (Student)	82.91 ± 0.11	58.52 ± 0.09	79.19 ± 0.43	60.00 ± 0.12	88.10 ± 0.18	34.01 ± 0.90	31.54 ± 0.39	23.84 ± 0.25	3.53 ± 0.05
ADASYN	APEX (Teacher)	87.06 ± 0.22	60.15 ± 0.10	77.60 ± 0.48	63.07 ± 0.15	88.06 ± 0.10	42.30 ± 0.63	33.47 ± 0.37	28.10 ± 0.26	27.80 ± 0.14
	DEFT (Student)	86.04 ± 0.17	59.72 ± 0.18	78.31 ± 0.66	62.35 ± 0.26	88.67 ± 0.25	41.96 ± 0.44	33.18 ± 0.68	27.10 ± 0.51	3.87 ± 0.04

Table 3 Five-Fold Cross-Validation Results of Teacher-Student Models on Taiwan Dataset Across Sampling Methods

Sampling	Model	Fold	Accuracy	Precision	Recall	F1	AUC	PR AUC	MCC	Cohen Kappa	Train Time (sec)
SMOTE	APEX (Teacher)	1	95.09	67.73	83.18	72.74	89.91	55.75	48.51	45.76	10.1106
		2	94.57	66.34	82.92	71.37	90.95	52.87	46.38	43.11	10.0150
		3	94.43	65.75	81.74	70.59	89.29	47.28	44.72	41.57	10.5055
		4	93.91	64.36	80.38	68.96	91.52	54.57	41.77	38.40	10.1910
		5	93.99	64.51	80.42	69.13	90.78	53.93	42.02	38.72	10.5574
	DEFT (Student)	1	93.70	63.91	80.27	68.47	93.16	50.91	41.03	37.48	1.6568
		2	93.99	64.75	81.52	69.54	93.76	48.10	43.13	39.56	1.5261
		3	93.70	64.15	81.36	68.88	93.14	55.05	42.13	38.32	1.5013
		4	93.84	64.68	82.54	69.61	92.84	50.18	43.71	39.75	1.5412
		5	93.62	64.01	81.33	68.72	92.97	49.54	41.90	38.01	1.5467
SMOTEENN	APEX (Teacher)	1	91.20	61.62	85.57	66.06	91.48	36.32	40.65	33.55	9.4993
		2	91.13	61.31	84.43	65.61	91.37	39.62	39.47	32.64	9.4534
		3	91.57	62.04	85.76	66.66	90.65	38.79	41.50	34.62	9.4700
		4	91.20	61.40	84.47	65.72	90.40	38.02	39.64	32.85	9.3412
		5	91.50	61.96	85.72	66.54	89.49	44.97	41.33	34.40	9.3267
	DEFT (Student)	1	90.25	60.63	85.08	64.61	93.77	43.03	38.62	31.00	1.4523
		2	90.18	60.78	86.14	64.82	92.88	45.84	39.47	31.47	1.4638
		3	90.25	60.85	86.17	64.93	92.97	42.03	39.62	31.66	1.2532
		4	90.98	61.37	85.45	65.71	93.43	38.10	40.16	32.93	1.5403
		5	90.91	61.51	86.52	65.92	92.85	43.46	41.01	33.40	1.2974
ADASYN	APEX (Teacher)	1	94.79	66.69	81.93	71.53	90.76	53.88	46.18	43.38	10.3351
		2	93.55	63.62	80.19	68.16	91.34	48.20	40.56	36.88	10.2155
		3	94.13	65.07	81.59	69.88	88.20	47.71	43.64	40.21	10.2084
		4	94.06	64.67	80.45	69.30	90.25	54.25	42.28	39.04	10.2450
		5	93.99	64.51	80.42	69.13	90.24	53.03	42.02	38.72	10.4852
	DEFT (Student)	1	93.55	64.11	82.39	68.96	93.41	50.54	42.75	38.53	1.5683
		2	93.11	63.08	81.06	67.65	92.90	49.40	40.31	36.01	1.6309
		3	93.33	63.70	82.27	68.49	92.59	52.44	42.06	37.66	1.5948
		4	92.96	62.58	79.89	66.97	92.83	51.36	38.79	34.65	1.5493
		5	93.77	64.30	81.40	69.04	93.11	51.79	42.38	38.62	1.5431

4.5 Fold-level performance consistency for Poland dataset

Table 4 presents fold-level performance for the Poland dataset across the different sampling approaches. APEX teacher recall achieves 78.34% accuracy in the fourth fold of SMOTE. The results of DEFT student recall reach 78.93% in the fourth SMOTE fold. The teacher recall in SMOTE-ENN averages at its maximum in the 5-fold at 79.78%. Student recall is strong and consistent at 79.78% in the first fold. ADASYN teacher recall fluctuates between 77.18% and 78.43%. Student models deliver similar recall within the range of 77.40% to 79.15%. For all configurations, the training time of student models averages 3.85 s. This demonstrates the successful transfer of minority class sensitivity from teacher to student.

4.6 Comparative analysis with classical and deep learning baselines

Table 5 shows the tremendous superiority of the proposed framework over seventeen baselines. Gradient Boosting has a high accuracy of 96.76%, and its recall is just 24.97%. HistGradientBoosting is proving to be similar with precision 57.52%. Deep learning models such as LSTM and BiLSTM have recall greater than 80% but a precision lower than 18%. APEX balanced precision and recall at 65.74% and 81.73% respectively. DEFT

Table 4 Five-Fold Cross-Validation Results of Teacher-Student Models on Poland Dataset Across Sampling Methods

Sampling	Model	Fold	Accuracy	Precision	Recall	F1	AUC	PR AUC	MCC	Cohen Kappa	Train Time (sec)
SMOTE	APEX (Teacher)	1	87.65	60.32	76.71	63.35	88.09	43.40	33.20	28.39	28.1349
		2	87.72	60.58	77.54	63.71	88.18	43.43	34.14	29.13	28.2776
		3	87.65	60.58	77.73	63.71	88.27	42.07	34.25	29.16	28.1460
		4	87.73	60.79	78.34	64.01	88.32	42.61	34.97	29.75	27.9859
		5	87.75	60.67	77.78	63.84	87.90	42.52	34.43	29.38	28.2060
	DEFT (Student)	1	86.94	60.29	78.49	63.26	88.80	41.51	34.25	28.56	3.9190
		2	87.04	60.23	77.97	63.19	88.57	41.85	33.84	28.35	3.8843
		3	86.49	60.02	78.48	62.82	88.68	40.06	33.78	27.87	3.9160
		4	86.70	60.25	78.93	63.17	88.99	40.85	34.44	28.50	3.9060
		5	86.80	59.96	77.51	62.79	88.67	41.59	33.11	27.64	3.8058
SMOTEENN	APEX (Teacher)	1	82.61	58.30	78.71	59.63	87.90	34.13	30.88	23.23	25.6999
		2	82.51	58.44	79.46	59.77	87.76	34.20	31.53	23.60	25.6144
		3	82.85	58.55	79.41	60.01	87.83	33.45	31.71	23.91	25.4824
		4	82.54	58.35	79.02	59.66	87.96	34.70	31.13	23.35	25.4484
		5	82.97	58.61	79.47	60.13	88.02	33.49	31.86	24.09	25.7254
	DEFT (Student)	1	82.92	58.66	79.78	60.17	88.40	34.41	32.11	24.22	3.4379
		2	82.86	58.50	79.19	59.96	87.94	32.57	31.50	23.79	3.5797
		3	82.77	58.43	79.02	59.84	88.05	34.18	31.29	23.59	3.5300
		4	82.93	58.56	79.33	60.06	88.12	34.99	31.69	23.96	3.5398
		5	83.08	58.45	78.62	59.96	88.00	33.90	31.11	23.67	3.5464
ADASYN	APEX (Teacher)	1	87.25	60.17	77.18	63.11	88.15	42.57	33.24	28.09	27.7414
		2	87.32	60.28	77.44	63.27	87.94	41.70	33.59	28.40	27.9315
		3	87.01	60.06	77.39	62.95	88.17	41.60	33.20	27.87	27.6228
		4	86.89	60.03	77.56	62.90	88.01	42.56	33.26	27.82	27.7597
		5	86.82	60.21	78.43	63.13	88.03	43.07	34.07	28.34	27.9409
	DEFT (Student)	1	86.15	59.56	77.40	62.15	88.36	41.50	32.37	26.62	3.8008
		2	86.22	59.91	78.68	62.64	88.66	41.99	33.72	27.62	3.8855
		3	86.04	59.92	79.15	62.62	89.05	42.63	34.02	27.68	3.8993
		4	85.99	59.65	78.11	62.24	88.69	41.64	32.93	26.89	3.9100
		5	85.79	59.56	78.22	62.09	88.61	42.05	32.86	26.69	3.8703

achieves this balance with accuracy achieved at 64.30% and recall is achieved at 81.40%. Training time for DEFT at 1.55 s is markedly superior compared to most of the baseline models. This well-balanced performance overcomes the critical deficiency of current methods.

It is critical to explicitly state a methodological caveat regarding the inclusion of sequence-based deep learning architectures such as RNN, LSTM, GRU, BiLSTM, and Transformer in this benchmark. Because these models possess inductive biases engineered for temporal or sequential dependencies, their application to static, non-temporal tabular financial data introduces an architectural misalignment. Consequently, as observed in our results, these models yield severely imbalanced performance profiles, characterized by artificially high recall but drastically diminished precision, reducing their viability as strict comparative baselines for tabular tasks.

To ensure a rigorously fair and highly competitive benchmarking environment, we have extensively augmented our comparative analysis with state-of-the-art algorithms explicitly optimized for heterogeneous tabular data. Specifically, we have integrated isolated configurations of LightGBM, CatBoost, XGBoost, and the attentive deep learning

Table 5 Comparative Evaluation of Classical Machine Learning, Deep Learning, and Knowledge Distillation Models on Taiwan Dataset

Model	Accuracy	Precision	Recall	F1	ROC AUC	Train Time (sec)
Gradient Boosting	96.76 ± 0.14	49.49 ± 4.34	24.97 ± 3.36	33.12 ± 3.55	92.49 ± 1.89	4.72 ± 0.08
HistGradientBoosting	96.90 ± 0.33	57.52 ± 12.31	22.17 ± 3.82	31.54 ± 4.34	92.61 ± 2.05	0.44 ± 0.02
Logistic Regression	86.91 ± 0.83	17.74 ± 1.49	84.06 ± 9.21	29.28 ± 2.47	91.61 ± 3.24	0.04 ± 0.01
KNN	96.90 ± 0.24	55.76 ± 10.79	19.37 ± 4.83	28.59 ± 6.35	82.21 ± 4.23	0.00 ± 0.00
AdaBoost	96.66 ± 0.26	48.37 ± 11.87	19.89 ± 4.52	27.58 ± 4.17	92.61 ± 1.72	0.96 ± 0.02
Decision Tree	95.29 ± 0.26	26.60 ± 4.16	26.11 ± 3.98	26.34 ± 4.00	61.85 ± 2.04	0.10 ± 0.01
Random Forest	96.79 ± 0.22	50.39 ± 12.11	12.51 ± 4.80	19.94 ± 7.05	91.33 ± 3.87	1.75 ± 0.07
Gaussian NB	73.47 ± 9.46	11.02 ± 3.68	91.44 ± 5.73	19.47 ± 5.63	90.64 ± 1.81	0.00 ± 0.00
Extra Trees	96.99 ± 0.19	70.67 ± 24.65	11.38 ± 6.42	19.13 ± 9.72	92.05 ± 2.57	0.80 ± 0.01
SVM (RBF)	96.76 ± 0.25	41.33 ± 30.06	5.11 ± 5.49	8.96 ± 9.29	91.09 ± 2.51	1.76 ± 0.12
RNN	87.11 ± 1.72	18.40 ± 2.72	85.78 ± 7.32	30.24 ± 3.86	93.07 ± 1.79	14.51 ± 6.24
LSTM	86.60 ± 3.51	17.94 ± 3.95	82.97 ± 5.28	29.32 ± 5.38	92.07 ± 2.54	33.50 ± 7.93
BiGRU	86.42 ± 1.86	17.08 ± 2.02	82.37 ± 9.38	28.24 ± 3.01	92.50 ± 2.35	27.38 ± 9.70
BiLSTM	86.07 ± 2.60	17.02 ± 2.61	83.02 ± 8.06	28.10 ± 3.27	92.24 ± 2.40	29.50 ± 8.66
GRU	85.30 ± 2.64	16.79 ± 3.27	86.94 ± 6.24	28.04 ± 4.60	92.95 ± 2.37	23.58 ± 11.14
Transformer	84.66 ± 3.93	16.20 ± 2.42	86.30 ± 8.21	27.14 ± 3.25	91.79 ± 2.12	44.88 ± 13.88
MLP	83.23 ± 2.67	14.64 ± 1.56	85.76 ± 9.72	24.95 ± 2.40	91.71 ± 2.92	7.66 ± 3.86
LightGBM	96.81 ± 0.42	50.98 ± 8.81	39.17 ± 8.92	44.00 ± 8.01	93.13 ± 2.95	0.69 ± 0.09
CatBoost	96.15 ± 0.43	41.29 ± 4.86	42.65 ± 5.63	41.67 ± 3.27	93.37 ± 1.97	13.20 ± 0.20
XGBoost	96.44 ± 0.49	44.55 ± 7.87	34.68 ± 3.38	38.78 ± 4.51	92.18 ± 2.17	0.66 ± 0.04
TabNet	83.46 ± 1.62	15.24 ± 2.06	89.73 ± 9.19	26.04 ± 3.32	93.35 ± 2.38	11.45 ± 4.42
APEX (Teacher)	94.40 ± 0.48	65.74 ± 1.39	81.73 ± 1.33	70.56 ± 1.58	90.49 ± 0.89	10.27 ± 0.24
DEFT (Student)	93.77 ± 0.15	64.30 ± 0.39	81.40 ± 0.81	69.05 ± 0.50	93.17 ± 0.36	1.55 ± 0.05

architecture TabNet. As demonstrated in the updated Tables 5 and 6, these tabular-native baselines perform significantly better than the sequence-based models, yet they still fall short of the balanced predictive fidelity and computational efficiency achieved by our proposed knowledge distillation framework. For instance, on the Taiwan dataset, the standalone LightGBM and CatBoost models achieve F1 scores of 44.00% and 41.67%, respectively, which are substantially outperformed by the APEX Teacher with an F1 score of 70.56% and the DEFT Student with an F1 score of 69.05%.

4.7 Robustness validation on Poland dataset

Table 6 confirms the robustness of the framework on a more complicated dataset of Poland. HistGradientBoosting has a high accuracy of 95.76% but low recall, at 18.49%. Logistic Regression has a recall of 61.89% and a precision of 11.11%. In recall, deep learning models range from 51.64% to 68.61%. Precision values for such models, however, are below 13% for all architectures. APEX has an accuracy of 60.59% and recall of 77.62%. DEFT is able to maintain precision to 60.15% and recall to 78.28%. Training time of DEFT at 3.89 s represents an 86% reduction compared to teacher training. This is the distilled student model that is showing how practical it can be deployed.

4.8 Synthesis of comparative findings

The proposed knowledge distillation framework delivers superior balanced predictive performance. Conventional classifiers achieve high accuracy but fail to detect minority class instances. Deep learning architectures provide stronger recall but suffer from low precision. APEX succeeds in striking the right balance between precision and recall across both datasets. DEFT sustains this balanced performance reduction while

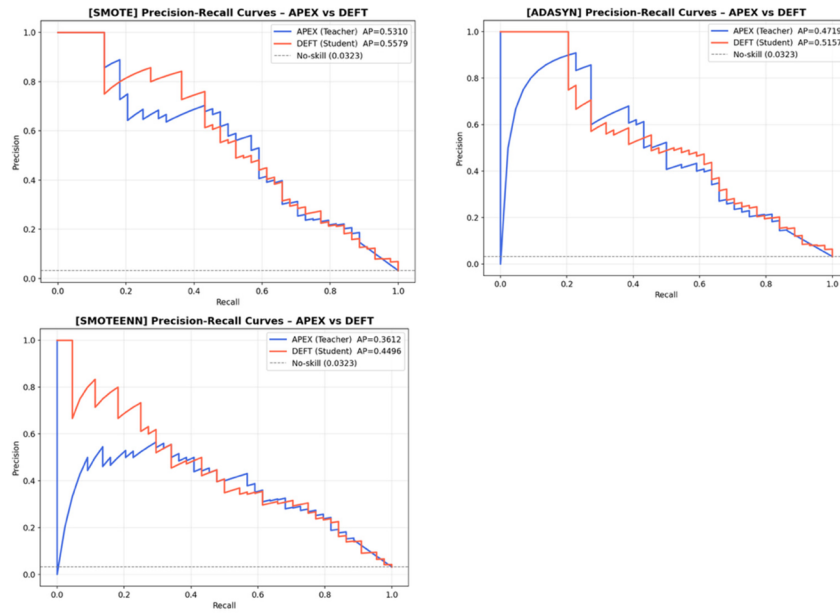
Table 6 Comparative Evaluation of Classical Machine Learning, Deep Learning, and Knowledge Distillation Models on Poland Dataset

Model	Accuracy	Precision	Recall	F1	ROC AUC	Train Time (sec)
HistGradientBoosting	95.76 ± 0.67	70.37 ± 11.63	18.49 ± 14.36	28.00 ± 17.67	86.92 ± 4.28	0.52 ± 0.11
Decision Tree	92.76 ± 1.11	25.42 ± 11.57	26.29 ± 13.35	25.84 ± 12.42	61.22 ± 6.91	1.11 ± 0.14
Logistic Regression	74.13 ± 1.16	11.11 ± 0.70	61.89 ± 3.12	18.83 ± 1.12	74.04 ± 1.96	1.19 ± 0.30
Gradient Boosting	95.43 ± 0.34	64.81 ± 10.91	10.21 ± 7.56	17.16 ± 11.03	82.54 ± 4.12	27.18 ± 0.37
Random Forest	95.27 ± 0.50	45.67 ± 20.46	7.99 ± 11.49	12.59 ± 16.33	84.80 ± 4.40	14.14 ± 0.86
Gaussian NB	5.96 ± 0.71	4.80 ± 0.01	97.78 ± 0.78	9.15 ± 0.01	52.26 ± 3.72	0.01 ± 0.00
AdaBoost	95.21 ± 0.12	12.90 ± 28.85	2.40 ± 5.37	4.05 ± 9.06	79.17 ± 3.96	5.30 ± 0.07
Extra Trees	94.72 ± 0.09	14.41 ± 3.17	1.86 ± 0.58	3.29 ± 0.96	78.46 ± 2.63	3.78 ± 0.12
KNN	95.02 ± 0.04	25.56 ± 9.94	1.74 ± 1.19	3.24 ± 2.17	65.10 ± 2.16	0.00 ± 0.00
SVM (RBF)	95.03 ± 0.05	27.21 ± 6.86	1.50 ± 0.37	2.84 ± 0.70	73.46 ± 1.14	214.02 ± 9.55
MLP	70.09 ± 5.31	10.56 ± 0.60	68.61 ± 10.98	18.23 ± 0.50	76.37 ± 1.31	43.55 ± 19.21
Transformer	76.57 ± 12.82	12.46 ± 3.71	51.64 ± 25.46	18.16 ± 1.58	74.30 ± 1.95	126.60 ± 40.47
BiLSTM	71.96 ± 10.22	11.16 ± 2.76	61.88 ± 21.61	17.84 ± 1.20	74.19 ± 3.26	169.63 ± 120.97
BiGRU	70.87 ± 5.83	10.40 ± 0.86	64.94 ± 12.93	17.80 ± 1.07	74.52 ± 3.55	174.89 ± 116.49
GRU	68.80 ± 6.91	10.07 ± 0.58	67.77 ± 14.89	17.40 ± 0.45	74.90 ± 2.99	246.38 ± 74.47
LSTM	73.39 ± 9.53	10.53 ± 1.39	56.97 ± 19.46	17.31 ± 1.03	73.60 ± 3.60	165.80 ± 115.15
RNN	71.52 ± 8.55	10.35 ± 1.53	60.57 ± 15.71	17.29 ± 0.97	73.01 ± 1.35	43.82 ± 13.13
CatBoost	91.77 ± 1.50	31.73 ± 7.27	58.59 ± 9.17	41.11 ± 8.25	87.20 ± 4.24	6.74 ± 0.10
XGBoost	93.47 ± 1.11	35.80 ± 9.34	44.06 ± 11.85	39.50 ± 10.45	85.63 ± 4.64	0.42 ± 0.02
LightGBM	87.98 ± 2.06	23.89 ± 5.13	65.43 ± 7.03	34.91 ± 6.32	86.86 ± 4.20	0.34 ± 0.03
TabNet	89.04 ± 3.86	16.22 ± 3.41	26.84 ± 9.92	19.13 ± 1.50	73.13 ± 0.92	183.82 ± 65.01
APEX (Teacher)	87.70 ± 0.05	60.59 ± 0.17	77.62 ± 0.59	63.72 ± 0.24	88.15 ± 0.17	28.15 ± 0.11
DEFT (Student)	86.79 ± 0.21	60.15 ± 0.15	78.28 ± 0.55	63.05 ± 0.22	88.74 ± 0.16	3.89 ± 0.05

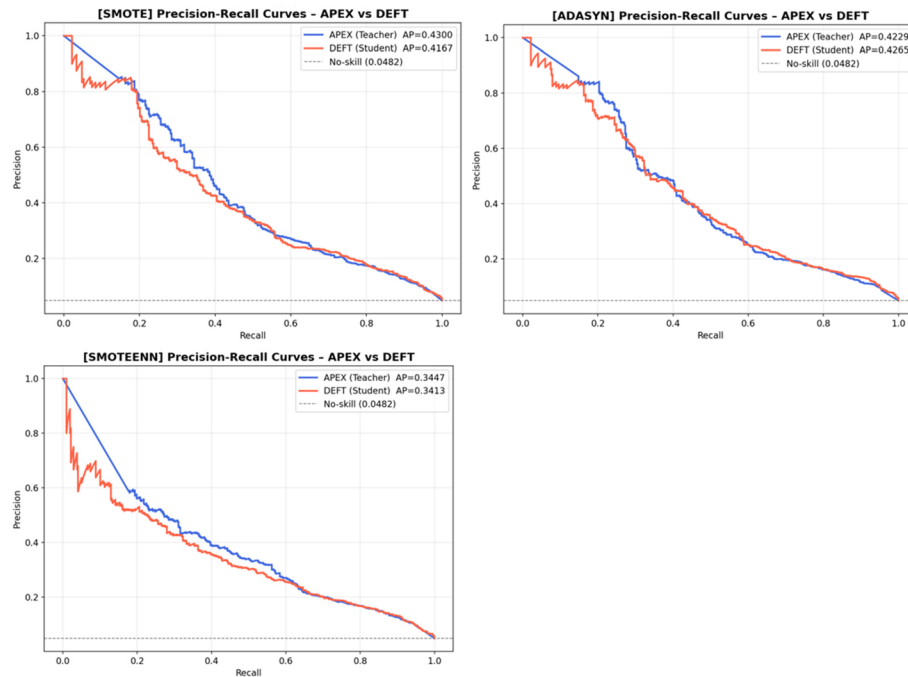
substantially mitigating computational overhead. The Taiwan dataset and Poland dataset compression of training time are over 80% and 86% respectively. Such efficiency improvement justifies the usefulness of the knowledge distillation mechanism both as a research and deployment tool. The framework directly solves one of the critical failures of the current practices on predicting imbalanced bankruptcies. This close-to-accurate detection ability is also supported by the precision-recall curves in Fig. 5, the threshold-free discrimination in the ROC curves in Fig. 6, and the latent representations and decision boundaries visualized through the high-dimensional t-SNE manifold projections in Fig. 7.

4.9 Statistical significance and model confidence set analysis

While aggregate metrics obtained from the 5-fold cross-validation protocol demonstrate the strong empirical performance of the proposed framework, rigorous statistical validation is required to substantiate claims of superior predictive fidelity. To determine whether the DEFT student model significantly outperforms competing approaches, it was compared against the three highest-performing baseline architectures: LightGBM, XGBoost, and CatBoost. Statistical evaluation was conducted using paired Student's t-tests and Wilcoxon signed-rank tests applied to the cross-validation F1-scores. In addition, the Model Confidence Set method proposed by Peter Reinhard Hansen et al. [38] was employed to establish definitive superiority among competing forecasts. The MCS procedure iteratively applies an equivalence test and removes models exhibiting significantly inferior performance until only a set of statistically indistinguishable superior models remains.



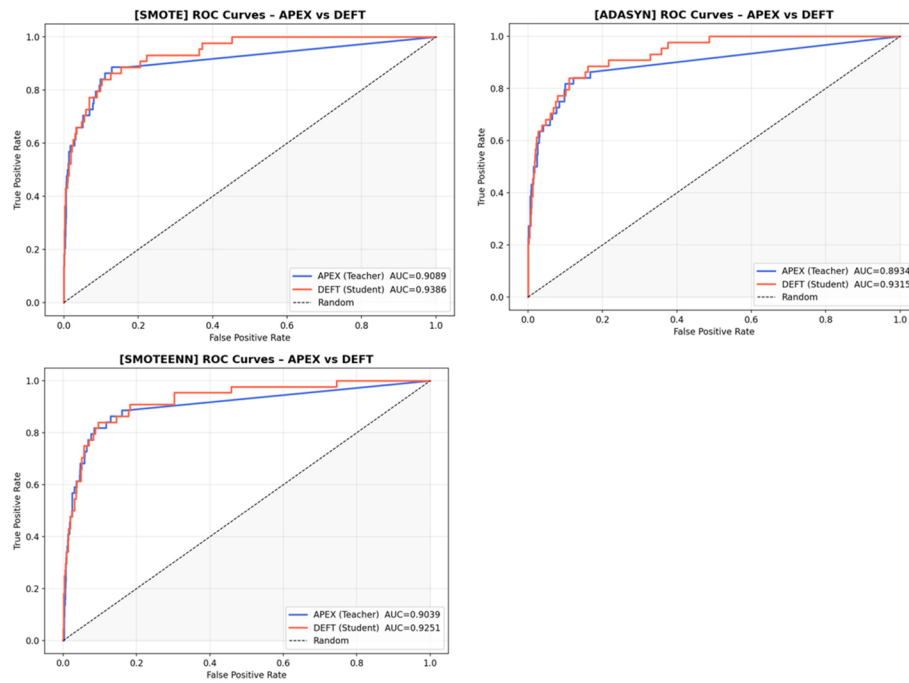
(a) Precision-Recall Curve - APEX vs DEFT in Taiwan Dataset



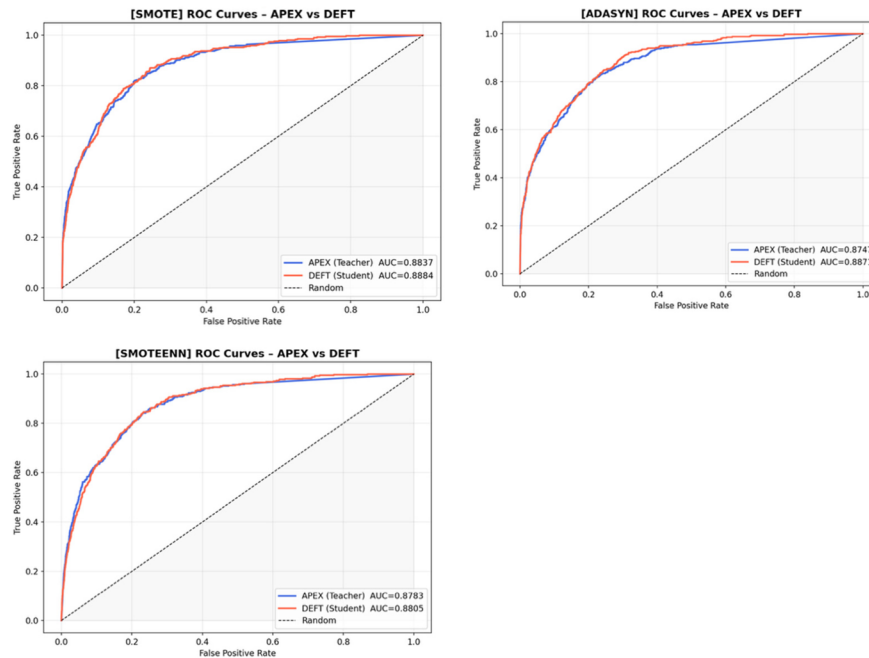
(b) Precision-Recall Curve - APEX vs DEFT in Poland Dataset

Fig. 5 Precision-Recall Trajectories Demonstrating Minority Class Detection Capability and Robustness of Teacher and Student Models

As summarized in Table 7, the paired t-tests produced highly significant p-values for all three comparisons, leading to rejection of the null hypothesis of equal predictive performance and confirming the statistical superiority of DEFT. Owing to the discrete nature of the 5-fold cross-validation protocol, which provides only five paired observations, the minimum attainable two-sided Wilcoxon p-value is 0.0625. The DEFT model



(a) ROC Curve APEX vs DEFT in Taiwan Dataset

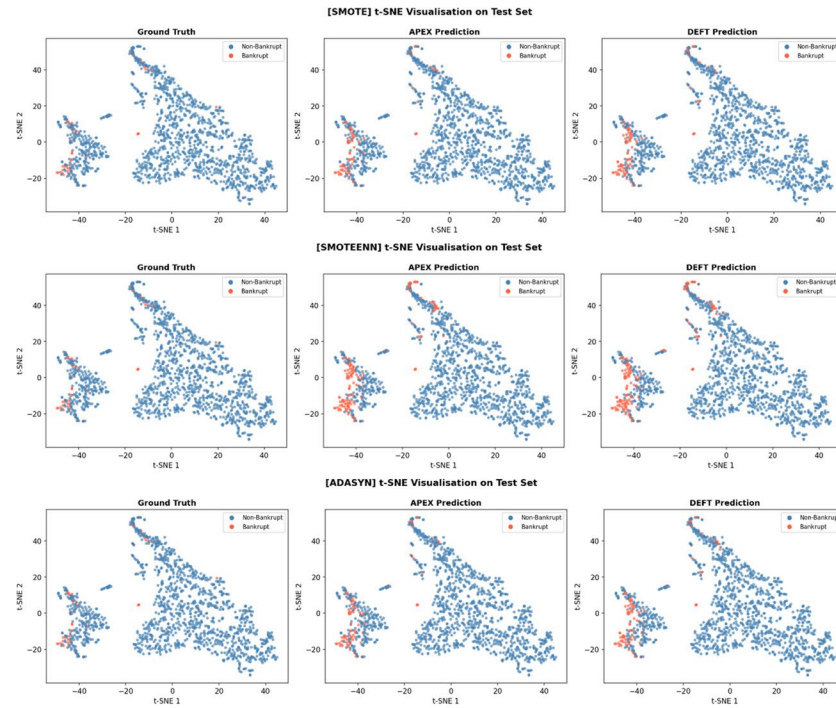


(b) ROC Curve APEX vs DEFT in Poland Dataset

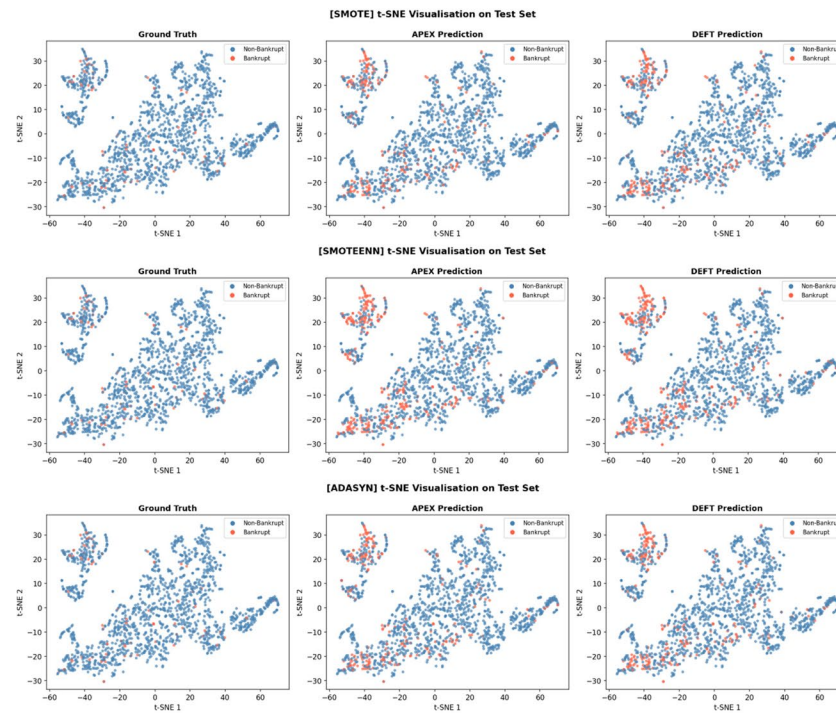
Fig. 6 Receiver Operating Characteristic Curves Evaluating Threshold-Invariant Discrimination Across Diverse Resampling Strategies

achieved this minimum value against all competing baselines, providing the strongest possible evidence of superiority under this nonparametric testing framework.

The MCS analysis further reinforced these findings. The procedure sequentially eliminated all baseline models, producing MCS p-values of 0.0 for LightGBM, 0.0 for CatBoost, and 0.0 for XGBoost. In contrast, the DEFT student model remained the only



(a) t-SNE Visualization on Test Set in Taiwan Dataset



(b) t-SNE Visualization on Test Set in Poland Dataset

Fig. 7 High-Dimensional t-SNE Manifold Projections Depicting Latent Representations and Decision Boundaries of Teacher-Student Networks

Table 7 Paired Statistical Significance Tests Evaluating DEFT (Student) Superiority

Model Compared	Wilcoxon p -value	Paired t-test p -value	DEFT Superiority Confirmed?
LightGBM	6.2500e-02	9.0835e-06	Yes
CatBoost	6.2500e-02	5.9678e-06	Yes
XGBoost	6.2500e-02	1.3417e-04	Yes

Table 8 Hyperparameter Sensitivity Analysis for Knowledge Distillation (T and α)

Model	Accuracy	Precision	Recall	F1	ROC AUC
DEFT ($T=3.0$, $\alpha=0.7$)	93.77 $\pm$ 0.15	64.30 $\pm$ 0.39	81.40 $\pm$ 0.81	69.05 $\pm$ 0.50	93.17 $\pm$ 0.36
DEFT ($T=2.0$, $\alpha=0.5$)	93.55 $\pm$ 0.44	64.00 $\pm$ 0.99	81.73 $\pm$ 1.37	68.75 $\pm$ 1.22	89.36 $\pm$ 1.09
DEFT ($T=5.0$, $\alpha=0.5$)	93.59 $\pm$ 0.20	64.15 $\pm$ 0.52	82.19 $\pm$ 0.99	68.98 $\pm$ 0.68	89.37 $\pm$ 1.46
DEFT ($T=3.0$, $\alpha=0.3$)	93.52 $\pm$ 0.39	64.08 $\pm$ 0.79	82.37 $\pm$ 0.84	68.92 $\pm$ 0.93	90.33 $\pm$ 1.58

architecture retained within the Superior Confidence Set and achieved the maximum MCS p -value of 1.0. Collectively, these statistical analyses provide decisive evidence that the distilled framework delivers significantly superior predictive performance and robustness relative to the strongest competing methods.

4.10 Hyperparameter sensitivity analysis and reproducibility details

The effectiveness of the knowledge distillation process is strongly influenced by two key hyperparameters: the temperature scaling factor T , which regulates the smoothness of the teacher model's probability distribution, and the interpolation coefficient α , which determines the relative contribution of teacher-generated soft targets and ground-truth hard labels during training. To empirically validate the selected configuration of $T = 3.0$ and $\alpha = 0.7$, a sensitivity analysis was performed on the Taiwan dataset using SMOTE.

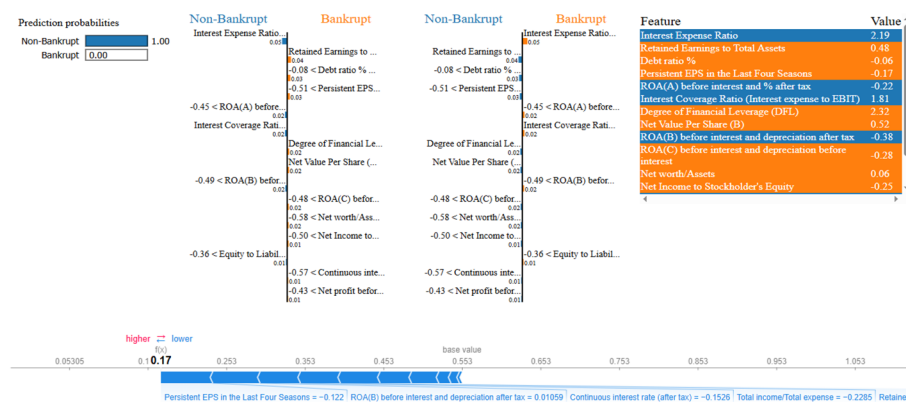
As presented in Table 8, variations in these hyperparameters produce measurable changes in the predictive behavior of the DEFT Student model. The configuration $T = 3.0$ and $\alpha = 0.7$ achieves the best overall performance, yielding the highest Accuracy of 93.77%, Precision of 64.30%, F1 Score of 69.05%, and ROC AUC of 93.17%. Reducing the interpolation coefficient to $\alpha = 0.3$, thereby increasing reliance on hard-label supervision, results in a modest improvement in Recall to 82.37% but simultaneously reduces ROC AUC and overall predictive balance. Likewise, increasing the temperature to $T = 5.0$ or decreasing it to $T = 2.0$ modifies the underlying class-probability structure learned from the teacher model without producing corresponding gains in predictive performance. Consequently, the configuration $T = 3.0$ and $\alpha = 0.7$ provides the most effective trade-off between knowledge transfer and decision-boundary preservation, enabling the student model to achieve superior overall generalization.

To ensure the rigorous reproducibility of the APEX-DEFT framework, all experimental procedures, hyperparameter optimizations, and evaluations were conducted within a highly controlled computational environment. The framework was implemented utilizing Python 3.10.x. Critical library dependencies and their exact versions are documented as follows: model development and classical machine learning baselines were executed using scikit-learn (v1.3.0); tree-based ensemble architectures relied on lightgbm (v4.0.0), xgboost (v2.0.0), and catboost (v1.2.0); and deep learning architectures were constructed using pytorch (v2.0.1). Furthermore, data resampling protocols were strictly managed using imbalanced-learn (v0.11.0), and post-hoc model interpretability was generated via shap (v0.42.1) and lime (v0.2.0.1). To guarantee accessible replication of our

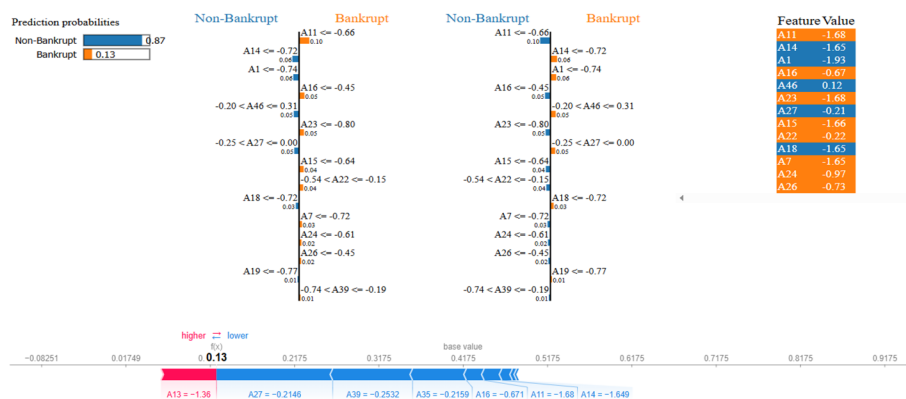
computational training times and latency metrics, all model training and evaluation paradigms were executed within a standard Kaggle notebook environment equipped with an NVIDIA Tesla P100 GPU.

5 Model explainability and interpretability framework

In high-stakes financial fields like predicting bankruptcy the use of unexplainable, black box models is fundamentally untenable given that transparency, accountability and regulatory adherence are paramount. Predictive decisions in these contexts have a direct impact on financial interventions, allocation of credit and risk management strategies, and, thus, require rigorous mechanisms of interpretability. To meet the latter need, the framework to be proposed exhibits the model-agnostic and model-specific interpretability measures in a complementary way that not only explains the global model behaviour but also local model behaviour. Specifically, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) are utilized to present a global feature contribution knowledge at the aggregate and instance specific level. Localized interpretability diagnostics integrating these SHAP and LIME attributions for instance-level predictions are presented in Fig. 8.



(a) LIME and SHAP Explanation in Taiwan Dataset using SMOTE



(b) LIME and SHAP Explanation in Poland Dataset using SMOTE

Fig. 8 Localized Interpretability Diagnostics Integrating Model-Agnostic LIME and SHAP Attributions for Instance-Level Predictions

5.1 Shapley additive explanations (SHAP)

SHAP is based on cooperative game theory and gives each feature a contribution value that relates to its marginal contribution for a certain prediction. The method calculates Shapley values that indicate an equitable distribution of prediction result among all input features by taking into account all possible subsets of features. Formally, the Shapley value ϕ_i associated with feature i for a model f and input instance x is defined as

$$\phi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)] \quad (28)$$

Here, F corresponds to the full feature set and S is the attribute with feature i not included during the evaluation process. This guarantees additive attribution by striving at contributions of all possible coalitions of features. For APEX Teacher using LightGBM, TreeExplainer is an implementation of computing exact Shapley values using tree traversal. This makes it less computationally complex than exhaustive enumeration, without a corresponding loss in precision of attribution. On the other hand, DEFT Student requires model agnostic Kernel Explainer for Shapley approximation using local regression. Perturbed samples and clustering based background distributions make it possible to achieve tractable estimation of the contributions of features. The global SHAP beeswarm visualizations quantifying these feature attributions and marginal impact distributions across target classes are shown in Fig. 9.

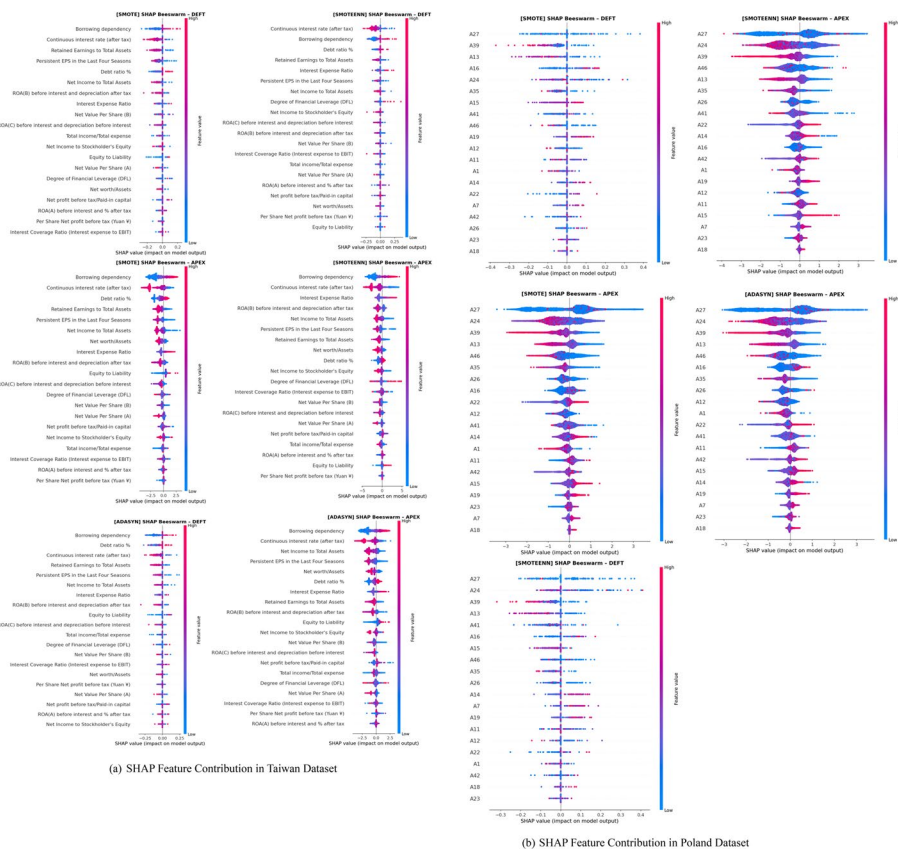


Fig. 9 Global SHAP Beeswarm Visualizations Quantifying Feature Attribution and Marginal Impact Distribution Across Target Classes

5.2 Local Interpretable Model-agnostic Explanations (LIME)

To further verify the explanations obtained from SHAP, LIME is used as an additional interpretability mechanism based on local fidelity. LIME builds an interpretable surrogate model that approximates the behavior of a complex model that is non-linear around a particular instance. By this approach individual predictions are easily understood intuitively by converting them into a simplified linear representation. The optimization objective of LIME is given by

$$\xi(x) = \arg \min_{g \in G} [L(f, g, \pi_x) + \Omega(g)] \quad (29)$$

where f denotes the original complex model, g is the interpretable surrogate model drawn from a class of simple models G , and $L(f, g, \pi_x)$ quantifies the lack of fidelity between f and g within a local neighborhood of the instance x . The term $\Omega(g)$ penalizes model complexity to enforce interpretability. The locality of the approximation is governed by a proximity weighting function $\pi_x(z)$, defined as

$$\pi_x(z) = \exp\left(-\frac{D(x, z)^2}{\sigma^2}\right) \quad (30)$$

Here, $D(x, z)$ is distance while sigma controls locality kernel width. Samples that are closer have more effect on the training of a surrogate model. Perturbations and discretization Opening the investigation Local explanation LIME is able to offer instance specific explanations that complements SHAP attributions. Granular LIME feature weight contributions elucidating these local surrogate explanations for individual bankruptcy predictions are illustrated in Fig. 10.

5.3 Limitations and future works

Although the predictive fidelity and exponential computational acceleration offered by the APEX-DEFT knowledge distillation framework is exceptional, there is a minor rigid scaling parameter of temperature. The current architecture uses a set distillation hyperparameter configuration that is predetermined and it could potentially mildly limit optimal knowledge transfer in circumstances of sudden and unprecedented macroeconomic volatility. To overcome this small drawback, the propose to investigate hyperparameter optimization algorithms based on dynamic reinforcement learning. Integrating adaptive temperature scaling mechanisms would enable the student network to autonomously calibrate its soft target assimilation, thereby providing consistently superior student generalization and alignment to the teacher ensemble across rapidly evolving financial markets.

While the combination of SHAP and LIME gives us powerful and model agnostic interpretability needed for regulatory purposes, computational overhead of creating this post-hoc explanations to vast data streams is a marginal constraint. Generating precise Shapley values by definition requires local perturbations that cause minor reduction in the instant inference capability of the lightweight DEFT student model during the ultra large batch processing. In future, more importance should be placed on development of inherently interpretable student architectures, or implementation of accelerated neural network based surrogate explainers. This advancement would enable decision-making

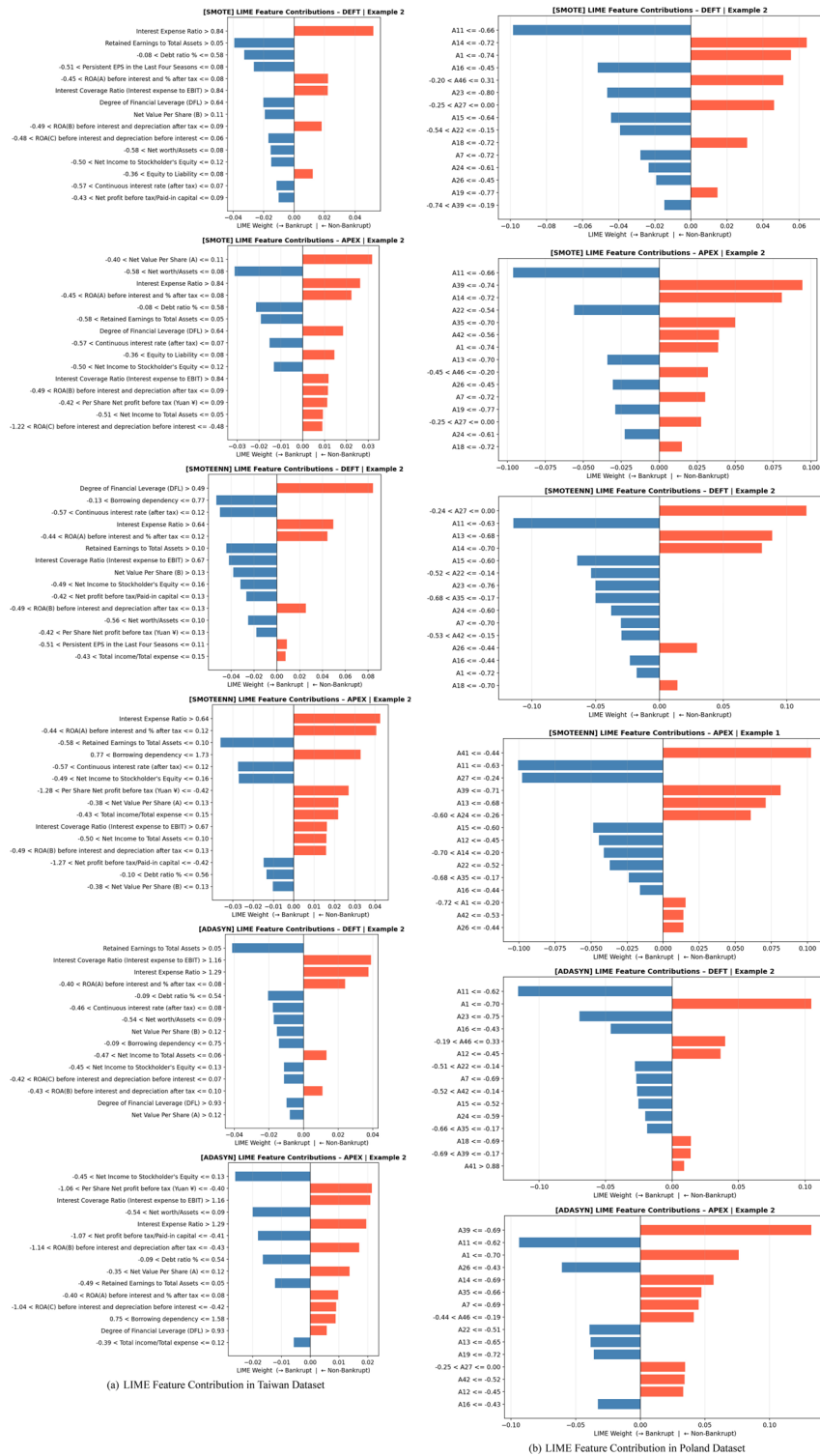


Fig. 10 Granular LIME Feature Weight Contributions Elucidating Local Surrogate Explanations for Individual Bankruptcy Predictions

at lightning speed, strengthening the trust of the stakeholders without compromising on the set up computational efficiency.

An additional limitation of the current empirical evaluation arises from its geographical and sectoral concentration. The Taiwanese and Polish datasets used in this study, while providing strong benchmarks for extreme class imbalance, are primarily composed of publicly listed companies and manufacturing-oriented enterprises. Therefore, a key direction for future research involves systematically assessing the extensibility of the APEX-DEFT framework across broader macroeconomic settings and heterogeneous regulatory environments. Subsequent studies should validate its generalization capacity using cross-regional datasets spanning both emerging and developed economies, as well as across diverse industrial domains including technology, services, and healthcare. Extending the analysis to unlisted private firms and small and medium sized enterprises is also essential, particularly given the increased noise and reduced standardization in their financial reporting structures compared to publicly traded corporations. Furthermore, while the current APEX-DEFT methodology is grounded exclusively in structured financial ratios, a significant research frontier lies in the adoption of multimodal learning paradigms. Future work should investigate the incorporation of unstructured textual sources such as earnings call transcripts and financial reports through dual encoder architectures and multimodal fusion strategies, where narrative disclosures may provide complementary informational signals that enhance the predictive fidelity of the distilled student network and improve its robustness across complex financial environments.

5.4 Conclusions

The accurate anticipation of corporate bankruptcy extends far beyond a standard classification challenge; it is a fundamental pillar of systemic economic stability, institutional capital adequacy, and proactive risk mitigation. By successfully distilling the complex predictive representations of a high-capacity ensemble into the lightweight DEFT student network, this research delivers profound financial implications for institutional risk management. The proposed framework equips financial institutions, credit rating agencies, and regulatory bodies with a highly precise early-warning system capable of navigating extreme class imbalances. This enables stakeholders to transition from reactive crisis management to proactive asset reallocation, hedging, and credit portfolio optimization, thereby mitigating the systemic shockwaves traditionally associated with corporate defaults.

Beyond its predictive fidelity, the true value of the APEX-DEFT framework lies in its exceptional viability for real-world deployment. In high-frequency trading and large-scale credit scoring environments, computational latency is a critical bottleneck. By reducing inference and training latency by over 80%, the distilled student model facilitates real-time risk assessment on massive, streaming financial datasets. Furthermore, by strictly embedding SHAP and LIME algorithms into the deployment pipeline, the framework ensures that every algorithmic decision is accompanied by granular, instance-level transparency. This interpretability is not merely an academic asset but a strict prerequisite for deployment, ensuring that the model adheres to stringent regulatory compliance standards and maintains the trust of financial stakeholders.

Moving forward, several promising avenues for further research remain. First, while the current architecture is highly optimized for tabular data, future iterations must

embrace multimodal fusion, incorporating unstructured textual disclosures, such as earnings call transcripts and macroeconomic news sentiment, to capture holistic strategic risks. Second, to address sudden macroeconomic volatility, future frameworks should investigate dynamic, reinforcement learning-based hyperparameter optimization, allowing the distillation temperature to autonomously adapt to shifting market distributions. Finally, to eliminate the minor latency overhead associated with post-hoc explainers in massive data streams, subsequent research should prioritize the development of inherently interpretable student architectures, continuing to advance the methodological frontier of scalable, transparent corporate risk assessment.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Grammarly in order to proofread. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Registry, trial registration number, and data of registration

Not applicable.

Authors' contributions

Abdul Kadar Muhammad Masum: Conceptualization, Methodology, Software, Formal analysis, Data Curation, Writing – Original Draft, Visualization. Md. Abul Kalam Azad: Conceptualization, Methodology, Validation, Investigation, Resources, Writing – Review & Editing, Supervision. Md. Abdur Rahman: Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft. Md. Tofael Ahmed Bhuiyan: Software, Validation, Formal analysis, Investigation, Data Curation, Visualization. Thanh Lam Nguyen: Conceptualization, Validation, Resources, Writing – Review & Editing, Supervision. Hasan Dinçer: Conceptualization, Methodology, Formal analysis, Writing – Review & Editing, Supervision. Serhat Yuksek: Conceptualization, Methodology, Formal analysis, Writing – Review & Editing, Supervision.

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Data availability

The datasets and code that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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